

Kinematics

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1 Introduction

A central topic in physics is understanding the motion of a particle. We would like to know the position, velocity, speed, acceleration, jerk, and other quantities relating to the motion of a particle. In several cases, the acceleration of a particle is zero, such as when the net force is zero. In these situations, it is very convenient to use the kinematic equations which we derive in this handout.

2 Preliminaries

We start with particle motion. The position of a particle, $x(t)$ is its position in space. We commonly write it as a function of time, t . In three dimensions, x can be a vector with 3 components. From this, we can obtain two more standard definitions.

Definition 2.1: The velocity of a particle, $v(t)$ is equal to $\frac{d}{dt}x(t)$.

Definition 2.2: The acceleration of a particle, $a(t)$ is equal to $\frac{dv}{dt} = \frac{d^2}{dt^2}x(t)$.

We say that a particle is “speeding up” if its velocity and acceleration vectors have the same sign and that a particle is “slowing down” otherwise. From these definitions, it is clear that we can go “backwards” via the fundamental theorem of calculus.

Theorem 2.1: Given initial position $x_0 = x(0)$,

$$x(t) = \int_0^t v(t) dt + x_0.$$

Proof. We have by the fundamental theorem of calculus

$$\int_0^t v(t) dt + x_0 = \int_0^t \frac{dx}{dt} dt + x_0 = \int_0^t dx + v_0 = x(t) - x(0) + x_0 = x(t).$$

□

The above theorem shows how to get the position from velocity.

Example 2.1: A particle is at position $x = 5$ at $t = 0$ and its velocity function can be modeled as $v(t) = t^3$. We can find the position function using the prior theorem as follows:

$$x(t) = \int_0^t v(t) dt + x_0 = \int_0^t t^3 dt + 5 = \frac{t^4}{4} + 5.$$

We can similarly obtain an expression for velocity in terms of acceleration.

Theorem 2.2: Given initial position $v_0 = v(0)$,

$$v(t) = \int_0^t a(t) dt + v_0.$$

Proof. We have by the fundamental theorem of calculus

$$\int_0^t a(t) dt + v_0 = \int_0^t \frac{dv}{dt} dt + v_0 = \int_0^t dv + v_0 = v(t) - v(0) + v_0 = v(t).$$

□

3 Derivation of the Kinematic Equations

Our goal in this handout is to derive the 4 standard kinematic equations when assuming constant acceleration. We start with the assumption

$$a = c.$$

Rewriting this gives

$$\frac{dv(t)}{dt} = c$$

and integrating once gives

$$v(t) = ct + v(0).$$

This is the first kinematic equation. It tells us that when a particle starts from rest (there is no initial velocity when starting from rest), the velocity of the particle is proportional to the time. We can write the constant acceleration c , as a to obtain the following canonical equation.

Kinematic Equation 1 For a particle undergoing constant acceleration a ,

$$v(t) = at + v_0.$$

We can now rewrite this as

$$\frac{dx(t)}{dt} = ct + v(0)$$

and integrating once gives

$$x(t) = \frac{c}{2}t^2 + v(0)t + x(0).$$

Understanding that a is constant, we can write c as a to get the canonical first kinematic equation that tells us that when a particle is undergoing constant acceleration, its position is proportional to the time squared when it starts from rest at position $x = 0$.

Kinematic Equation 2 For a particle undergoing constant acceleration a ,

$$x(t) = \frac{1}{2}at^2 + v_0t + x(0).$$

From the previous two equations, we can easily derive the third kinematic equation.

Kinematic Equation 3 For a particle undergoing constant acceleration,

$$x(t) - x(0) = \left(\frac{v_0 + v(t)}{2} \right) t.$$

Proof. We can use Theorem 2.1 to get

$$\left(\frac{v_0 + v(t)}{2} \right) t = \left(\frac{at + v_0 + v_0}{2} \right) t = \frac{at^2 + 2v_0t}{2} = \frac{1}{2}at^2 + v_0t.$$

We know that this is equal to $x(t) - x(0)$ by Theorem 2.2. □

The final kinematic equation relates the quantities of position, velocity, and acceleration without mentioning time.

Kinematic Equation 4 For a particle undergoing constant acceleration,

$$v(t)^2 - v(0)^2 = 2a(x(t) - x(0)).$$

Proof. We can use Theorem 2.1 to isolate an expression for a as

$$a = \frac{v(t) - v(0)}{t}.$$

We can now use Theorem 2.3 from above to simplify the right-hand side as

$$2a(x(t) - x(0)) = 2 \cdot \frac{v(t) - v(0)}{t} \cdot (t) \frac{v(t) + v(0)}{2}$$

and cancelling out the $2t$ on the numerator and denominator gives

$$2a(x(t) - x(0)) = (v(t) - v(0))(v(t) + v(0)) = v(t)^2 - v(0)^2$$

as desired. □

Remark 3.1: The theorems above are only valid when a particle is going under constant acceleration and be false otherwise.

Remark 3.2: The $t = 0$ does not have to be the absolute start time. It may be more convenient in some problems to label $t = 0$ after some event occurs that causes the particle under consideration to have constant acceleration.

Remark 3.3: All of the theorems can be derived by the fact that acceleration is the second derivative of position and then the constant acceleration twice. If you ever forget some of the equations, just remember the previous fact and then you can solve any kinematics problem via Theorem 2.1 and Theorem 2.2 (although there may be one additional step).

4 Examples

Example 1: A bug is originally moving to the right at a constant velocity of $2\frac{m}{s}$. After 10 seconds, a kid begins to push the bug and the bug begins to accelerate at $10\frac{m}{s^2}$ from here. What is the position of the bug after 50 seconds?

Solution: Let us consider the motion of the bug into two parts, before and after the kid intervened. Before the kid intervened, we had $x(t) = 2t \implies x(10) = 20m$. Now, for when the kid started pushing we have

$$x(t) = \frac{1}{2}10t^2 + 2t = 5t^2 + 2t \implies x(40) = 8080m.$$

Thus after 50 seconds, the bug is at position $8080 + 20 = 8100m$.

Example 2: A boy who is running at a constant velocity begins to accelerate at a constant rate of $0.50\frac{m}{s^2}$. After running for $20m$, his speed is $50\frac{m}{s}$. What was speed before accelerating?

Solution: We can use Theorem 2.4 to obtain

$$v(0)^2 = v(t)^2 - 2a(x(t) - x(0)) = 50^2 - 2(0.50)(20) = 2480\frac{m^2}{s^2} \implies v(0) = \sqrt{2480} \approx 49.8\frac{m}{s}.$$

Exponential Decay and Growth

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1 Introduction

In science, several types of behaviors can be modeled by curves that appear to increase or decrease very fast as well as curves that seem to be bounded above. For example, if you are modeling the amount of people living in a city, and a city can only hold 100 people, then the graph showing the city's population over time may look like a curve that increases very fast but then remains constant at 100. These are examples of exponential growth and decay, which we study in this handout.

2 The Derivatives of Exponentials

Before we can start talking about exponential growth and decay, we need to first discuss the most important topic: the derivative of e^x . What is e , you might wonder?

What is e ?

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

It is well known that

$$\frac{d}{dx}e^x = e^x$$

This means that the slope of the tangent line to e^x at any given x is equal to e^x . This beautiful result has several proofs and we give one now. We start with the definition of the derivative using a limit (refer to a previous handout if you are not fully comfortable with it or just skip this section and accept on faith that the derivative of e^x is e^x).

$$\frac{d}{dx}e^x = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = e^x \left(\frac{e^h - 1}{h} \right).$$

Now, define $n = e^h - 1$. This implies that $n + 1 = e^h$, which means that $h = \ln(n + 1)$. We can also compute

$$\lim_{h \rightarrow 0} n(h) = \lim_{h \rightarrow 0} e^h - 1 = 0.$$

So, when we substitute back in our new expressions in terms of n , we can replace the variable of the limit with n . We get

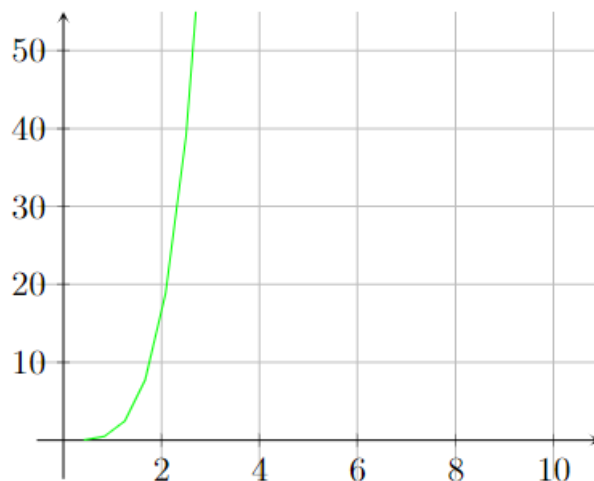
$$\frac{d}{dx}e^x = e^x \lim_{n \rightarrow 0} \frac{n}{\ln(n+1)} = e^x \lim_{n \rightarrow 0} \frac{1}{\frac{1}{n} \ln(n+1)} = e^x \lim_{n \rightarrow 0} \frac{1}{\ln((n+1)^{\frac{1}{n}})}.$$

We get the last equality by using the property of logarithms that $\ln(a^b) = b \ln(a)$. We can use the property of the limits of composite functions to put the limit inside of the logarithm and use our original definition for e to finally get

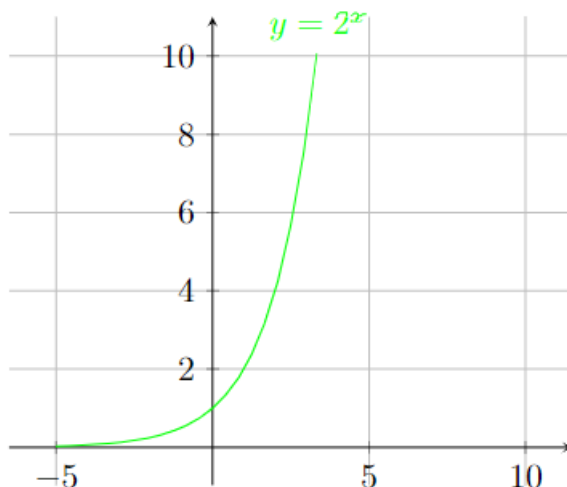
$$\frac{d}{dx}e^x = e^x \frac{1}{\ln(\lim_{n \rightarrow 0} (n+1)^{\frac{1}{n}})} = e^x \frac{1}{\ln(e)} = e^x.$$

3 Motivation

You have likely heard the phrase “exponential growth” several times before. However, it was probably used incorrectly most of the time. People tend to use the word “exponential” for something that seems to grow very large but this need not always be the case. For example, consider the following function.



This function seems to grow very fast, but it is actually not the graph of an “exponential” function. It is simply the graph of $f(x) = x^4$ which as we previously learned has derivative $f'(x) = 4x^3$ by the power rule. Exponential functions have the form $f(x) = c_1 e^{c_2 x}$. One example of an exponential function is depicted below.



This does not seem to be an exponential function, as one may wonder, where is the e ? To see that this is exponential, we use a key property of logarithms by rewriting this as $f(x) = 2^x = e^{\ln(2^x)} = e^{x \ln(2)}$ which fits the criteria outlined above. We can also find the derivative of this function as follows:

$$\frac{d}{dx} 2^x = \frac{d}{dx} e^{x \ln(2)} = e^{x \ln(2)} \cdot \ln(2) = 2^x \ln(2)$$

via the chain rule. We can see that $y'(x) = \ln(2)y(x) = ky(x)$. In other words, $y'(x) \propto y(x)$. Now, let us see if this holds in general for exponential functions.

$$\frac{d}{dx}c_1e^{c_2x} = c_1(e^{c_2x}c_2) = (c_1c_2)e^{c_2x}.$$

We see that $f'(x) = (c_1 \cdot c_2)f(x) = kf(x)$ where k is a constant and is set to $c_1 \cdot c_2$. We have derived that the derivative of an exponential function is proportional to the function itself, or in other words

$$f'(x) \propto f(x).$$

Conceptually, when k is positive, this means that as a function is taking on a larger value, it will start to increase more rapidly. This is known as exponential growth. Sometimes k is negative so $f'(x) = -kf(x)$ for positive k . For these functions, as the function takes on smaller values, it starts to decrease slower. This is known as exponential decay. The canonical example of exponential decay is **radioactive decay**.

4 Exponential Growth

Consider we are given that $f'(x) = kf(x)$ where k is a positive constant. From the last section, we can see that $f(x)$ is a function consisting of exponential growth. Our task becomes to derive $f(x)$. The first step is to rewrite this in Leibniz notation, as that makes the following steps clearer.

$$f'(x) = \frac{d}{dx}f(x) = kf(x).$$

Now, let $y = f(x)$. Then, we have

$$\frac{dy}{dx} = ky.$$

Separating variables gives

$$\frac{dy}{y} = kdx.$$

We integrate both sides to get

$$\int \frac{dy}{y} = \int kdx.$$

Using the fact that the derivative of $\ln(x)$ is x^{-1} and the power rule for integrals gives

$$\ln(y) = kx + c.$$

Remembering that exponentials are the inverse of logarithms, we can exponentiate each side to get

$$y = e^{kx+c}.$$

We can now use the key rule of exponents that $a^{b+c} = a^b a^c$ to rewrite this as

$$y = e^{kx} e^c$$

and substitute c_2 for k and c_1 for e^c to get this in the recognizable form

$$y = c_1 \cdot e^{c_2x}$$

which is the expected and appropriate function for this differential equation. Suppose we know the value of $y(0) = a$. Then, we have

$$c_1 \cdot e^0 = a \implies c_1 = a.$$

And, if also given the value of y at $x = b$, $f(b) = z$ then we have

$$a \cdot e^{c_2 b} = z \implies e^{c_2 b} = \frac{z}{a} \implies c_2 b = \ln\left(\frac{z}{a}\right) \implies c_2 = \ln\left(\frac{z}{a}\right)/b = \frac{1}{b}(\ln(z) - \ln(a)).$$

5 Exponential Decay

Consider k being negative (i.e. we are given that $f'(x) = -kf(x)$). We can write this using Leibniz notation as

$$\frac{df}{dx} = -kf(x)$$

and separate variables to get

$$\frac{df}{f(x)} = -k dx$$

and integrate to get

$$\int \frac{df}{f(x)} = \int -k dx.$$

We can compute the integrals to get

$$\ln(f(x)) = -kx + c_3.$$

We exponentiate to undo the logarithm to get

$$f(x) = e^{-kx+c_3} = e^{c_3} e^{-kx}.$$

We can see that e^{c_3} is a constant so we denote this as c_1 to get the final form

$$f(x) = c_1 e^{-kx}.$$

If we know the initial condition, i.e. $f(0) = a$, then we must have

$$c_1 e^{-k(0)} = a \implies c_1 = a.$$

6 Examples

Example 1: A student begins to spread a rumor at her school. Initially, only five students know about the rumor. Each student continuously tells the rumor to two other students in the school. Write a formula modeling the number of students in the school who know the rumor at a given time t .

Solution: By the result of Section 3, we have $c_1 = 5$. And, since if there are $f(t)$ students who know the rumor at any given time we have $f'(t) = 2f(t)$. Thus,

$$f(t) = 5e^{2t}.$$

Example 2: Consider the same example as before except we do not know how many students each student tells the rumor to but we do know that at time $t = 10$, $f(t) = 100$.

Solution: We have that $c_1 = 5$ as before. Now, we can use the results obtained at the end of Section 3 to get $c_2 = \frac{1}{10}(\ln(100) - \ln(5)) \approx 0.3$ so

$$f(t) = 5e^{0.3t}.$$

Example 3: Carbon-14 is always undergoing exponential decay with constant k . Carbon-14 has a half-life of approximately 5,730 years. This means that every 5,730 years the amount of Carbon-14 halves. If we have 50g of Carbon-14 today how much will be left after 100 years?

Solution: First, we derive a formula for half-life. If an isotope's exponential decay is modeled by $f(t) = f(0)e^{-kt}$ and has a half-life of z , then

$$f(a+z) = \frac{f(a)}{2} \implies f(0)e^{-ka-kz} = f(0)e^{-ka}e^{-kz} = f(0)\frac{e^{-ka}}{2} \implies e^{-kz} = \frac{1}{2}$$

so

$$-kz = \ln(0.5) \implies k = -\ln(0.5)/z.$$

Thus, in this example, $z = 5730$ so $k = -\ln(0.5)/5730 \approx 1.21 \cdot 10^{-4}$. Thus,

$$f(t) = 50e^{-1.21 \cdot 10^{-4} \cdot t} \implies f(100) \approx 49.3986g.$$

Separable Differential Equations

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1 Introduction

Separable differential equations are a type of nonlinear first order differential equations. They are first order differential equations as they only involve first derivative terms and are nonlinear since they may be, say, a quadratic function of the first derivative. Let y be a function of x , (i.e. $y = y(x)$).

Definition: A separable differential equation is of the form

$$y'(x) = \frac{f(x)}{g(y(x))}.$$

2 Derivation

We now derive the method for solving separable differential equations. Suppose we have a separable differential equation $y'(x) = f(x)/g(y(x))$. We can rewrite this using Leibniz notation as

$$\frac{dy}{dx} = \frac{f(x)}{g(y(x))}.$$

We can now multiply both sides by $g(y(x))$ and integrate with respect to x to get

$$\int g(y(x)) \frac{dy}{dx} dx = \int f(x) dx.$$

We integrate this using a substitution: $u = y(x)$. This implies that $du = y'(x) dx = \frac{dy}{dx} dx$. Plugging this in then gives

$$\int g(u) du = \int f(x) dx.$$

The differential equation thus reduces to a problem of integration. Although this is the mathematically correct way of solving the differential equation, there is an easier method that gives the same result. Given a separable differential equation, multiply both sides by $g(y(x)) dx$ to get

$$g(y(x)) dy = f(x) dx.$$

All that remains to be done is to integrate both sides,

$$\int g(y(x)) dy = \int f(x) dy.$$

This is how we will compute the examples.

3 Examples

Example 1: We solve the differential equation

$$\frac{dy}{dx} = \frac{x}{y}$$

using the method of separation of variables. In this example, $f(x) = x$ and $g(y(x)) = y$. We multiply both sides of the equation by $g(y(x)) dx = y dx$ to get $y dy = x dx$. We now integrate both sides:

$$\int y dy = \int x dx \implies \frac{y^2}{2} + c_1 = \frac{x^2}{2} + c_2.$$

We multiply both sides by 2 to get $y^2 + 2c_1 = x^2 + 2c_2$. Note that $2c_2 - 2c_1$ is also a constant, call it k . This reduces the prior equation to $y^2 = x^2 + k$ which is the simplified result.

Example 2: We solve the differential equation

$$\frac{dy}{dx} = \frac{\sin(x)}{\cos(y)}.$$

Multiply both sides by $\cos(y) dx$ to get $\cos(y) dy = \sin(x) dx$. Integrate both sides to get

$$\int \cos(y) dy = \int \sin(x) dx \implies \sin(y) + c_1 = -\cos(x) + c_2.$$

We can write $k = c_2 - c_1$ to give the solution $\sin(y) = k - \cos(x)$.

Example 3: Suppose that

$$\frac{dy}{dx} = y^2.$$

One might wonder how this can be a separable differential equation since there appears to be no $f(x)$ and we are multiplying by $g(y)$ rather than dividing by it. This is in fact separable since we can let $f(x) = 1$ and $g(x) = 1/y^2$. We can verify that $f(x)/g(y) = 1/(1/y^2) = y^2$. We multiply both sides of the differential equation by $g(y) dx = (1/y^2) dx$ to get

$$\frac{1}{y^2} dy = 1 dx$$

We can integrate both sides by using the power rule to get

$$\int \frac{1}{y^2} dy = \int 1 dx \implies -y^{-1} + c_1 = x + c_2.$$

We can let $k = c_2 - c_1$ to get $-y^{-1} = x + k$ as the solution.

Example 4: Let us compute one more example where we are given the initial conditions. The differential equation is

$$\frac{dy}{dx} = \frac{2x}{4y^3}$$

and $y(0) = 100$. We multiply both sides by $4y^3 dx$ to get $4y^3 dy = 2x dx$. Integrating both sides and applying the power rule gives

$$\int 4y^3 dy = \int 2x dx \implies y^4 + c_1 = x^2 + c_2.$$

We can write $k = c_2 - c_1$ to get $y^4 = x^2 + k$. Plugging in our initial condition forces $(100)^4 = 0^2 + k$ which implies that $k = \sqrt[4]{100} = \sqrt{10}$. Thus, the solution simplifies to $y^4 = x^2 + \sqrt{10}$.

Example 5: The most illuminating example is solving the differential equation from exponential growth:

$$\frac{dy}{dx} = ky.$$

This differential equation requires the knowledge of the derivatives of e^x and $\log(x)$ which is solved in the handout on exponential and logarithmic growth.

Second Order Differential Equations

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1 Introduction

Second-order linear differential equations are of the following form.

Second order Linear Differential Equations:

$$a(x)y''(x) + b(x)y'(x) + c(x)y(x) = d(x). \quad (1.1)$$

In this handout, we study **homogeneous equations**, where $d(x) = 0$. We require constant coefficients (i.e. $a(x), b(x), c(x)$ being constants). Several fundamental concepts in physics, such as Newton's equation and the Euler-Lagrange equations, are second-order linear differential equations. For example, the equation of motion for a system consisting of small oscillations in one dimension is $mx''(t) + kx(t) = 0$ which also has constant coefficients and is homogeneous. Thus, it is very important to know how to solve them.

2 The Solution

Consider an equation of the form 1.1 with constant coefficients and $d(x) = 0$. Let us write $a(x), b(x)$, and $c(x)$ as a, b , and c respectively. We reduce 1.1 into a simpler one by dividing both sides by a to get

$$y''(x) + \frac{b}{a}y'(x) + \frac{c}{a}y(x) = 0. \quad (2.1)$$

For simplicity, rename b/a as p and c/a as q . Thus, we need to solve

$$y''(x) + py'(x) + qy(x) = 0. \quad (2.2)$$

We use an **ansatz**, which is essentially a fancy glorified educated guess to solve the problem. Suppose that $y(x) = e^{cx}$ is a solution to the differential equation. Then, we have $y''(x) = c^2e^{cx}$ and $y'(x) = ce^{cx}$. Plugging this into 2.2 gives

$$c^2e^{cx} + p \cdot ce^{cx} + qe^{cx} = e^{cx}(c^2 + pc + q) = 0.$$

We can divide both sides by e^{cx} to get

$$c^2 + pc + q = 0.$$

This is called the **characteristic equation**. We can find c by factoring when possible and using the quadratic equation when it is not. The method from here depends on what type of roots we get and how many there are.

2.1 Two Real Roots

If there are two real roots, say, c_1, c_2 , then the general solution to the differential equation is $y(x) = Ae^{c_1x} + Be^{c_2x}$ where A, B can be any constants. We know from the ansatz that e^{c_1x} and e^{c_2x} work and it turns out that any linear combination of these terms is also a solution.

2.2 One Real Root

When there is one real root (i.e. the characteristic equation is of the form $(c - k)(c - k)$), then the general solution is of the form $y(x) = Ae^{kx} + Bxe^{kx}$.

2.3 Two Complex Roots

In this case, we get two complex roots z_1 and z_2 and plug them in the same way as if we had gotten two real roots to get a general solution of $y(x) = Ae^{z_1x} + Be^{z_2x}$. From here, we can use Euler's formula,

$$e^{ix} = \cos(x) + i\sin(x),$$

to simplify.

3 Examples

We compute one example of each type of solution.

Example 1: Solve the equation

$$2y''(x) + 16y'(x) + 30y(x) = 0.$$

Solution: We first divide both sides by 2 to get

$$y''(x) + 8y'(x) + 15y(x) = 0.$$

The characteristic equation we get is $(c^2 + 8c + 15) = 0$. We can factor the left hand side to get $(c + 5)(c + 3) = 0$. This implies that the solutions to the characteristic equation are $c = -5$ and $c = -3$. Thus, the general solution to the differential equation is

$$y(x) = Ae^{-5x} + Be^{-3x}.$$

Example 2: Solve the equation

$$y''(x) - 8y'(x) + 16y(x) = 0.$$

Solution: The characteristic equation is $(c^2 - 8c + 16) = 0$. We can factor the left hand side to get $(c - 4)(c - 4) = 0$ which implies that there is one real solution, namely $c = 4$. Thus, the general solution to the differential equation is

$$y(x) = Ae^{4x} + Bxe^{4x}.$$

Example 3: Solve the equation

$$y''(x) + 4y'(x) + 5y(x) = 0.$$

Solution: The characteristic equation is $(c^2 + 4x + 5) = 0$. We see that the discriminant of this equation is $4^2 - 4 \cdot 1 \cdot 5 < 0$ so the equation has two imaginary roots. Using the quadratic formula, we see that the roots of the equation are $z_1 = -2 + i$ and $z_2 = -2 - i$. So, the general solution to the differential equation is

$$y(x) = Ae^{(-2+i)x} + Be^{(-2-i)x}.$$

We can factor out e^{-2x} to get

$$y(x) = e^{-2x}(Ae^{ix} + B^{-ix}).$$

Applying Euler's formula to the terms in parenthesis gives

$$Ae^{ix} + B^{-ix} = A(\cos(x) + i \sin(x)) + B(\cos(-x) + i \sin(-x)).$$

We can use the trigonometric identities $\cos(-x) = \cos(x)$ and $\sin(-x) = -\sin(x)$ to get

$$Ae^{ix} + B^{-ix} = A(\cos(x) + i \sin(x)) + B(\cos(x) - i \sin(x)) = (A + B)(\cos(x)) + i(A - B)(\sin(x)).$$

We can see that $A + B$ and $A - B$ are also constants so we can substitute c_1 and c_2 for them respectively to get the general solution

$$y(x) = e^{-2x}(c_1 \cos(x) + ic_2 \sin(x)).$$

We leave the following example as an interesting, real-world occurring, exercise.

Example 4: Solve the differential equation for springs mentioned in the introduction. After you have done this, solve the differential equation for when damping is present: $mx''(t) + cx'(t) + kx(t) = 0$.

Gaussian Integrals

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1 Motivation: Probability Distributions

To motivate the importance of Gaussian integrals, consider their most important application to probability theory used in thermodynamics. Thermodynamics considers macroscopic systems involving a large number (on the order of 10^{23}) of atomic or molecular particles: too many to individually describe the mechanics of each individual particle. Fortunately, thermodynamics is often interested in “average” quantities of an entire system, rather than the specifics of individual molecules. Because there are so many particles in the macroscopic systems studied by thermodynamics, the specific values of quantities for individual molecules are often not relevant to the overall properties of a system, but rather their average values over many molecules. With this in mind, the specifics of individual particles can be ignored, and instead distribution for these quantities over the entire system is considered.

The distribution of a quantity can be described with a **probability density function**, or PDF. For a PDF $f(x)$, the $f(a)dx$ is equal to the probability of measuring x between the values of a and $a + dx$ in the distribution:

$$f(a)dx = \text{probability of finding } x \text{ between } a \text{ and } a + dx \quad (1.1)$$

The total probability of finding x within an interval $a < x < b$ is just the sum of $f(x)dx$ for each x within that interval. This summation is just the integral from a to b of $f(x)dx$:

$$\int_a^b f(x)dx = \text{probability of finding } x \text{ between } a \text{ and } b \quad (1.2)$$

Some of the most common PDFs found in thermodynamics and modern physics are the **Gaussian function**

$$f(x) = Ae^{-\alpha x^2} \quad (1.3)$$

and a related set of functions containing the Gaussian function multiplied by some polynomial

$$f(x) = Ax^n e^{-\alpha x^2}, \quad (1.4)$$

where A and a are parameters for the Gaussian function. A **Gaussian integral** is the definite integral of the Gaussian function from $-\infty$ to ∞ :

$$I = \int_{-\infty}^{\infty} Ae^{-\alpha x^2} dx \quad (1.5)$$

When finding probabilities with a Gaussian PDF, the need to compute Gaussian integrals invariably comes up as a result of Equation (1.2). This handout examines methods for computing Gaussian integrals and presents several practice problems relevant to physics Olympiads.

2 Gaussian Integrals

To start, consider the simple case of a Gaussian integral I with $A = \alpha = 1$ in Equation (1.5):

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx \quad (2.1)$$

Unfortunately, the general antiderivative of $e^{-x^2}dx$ can not be expressed in terms of elementary functions. However, the specific integral in Equation (2.1) can still be evaluated with a trick using polar coordinates.

Consider two Gaussian functions of two independent variables x and y . From basic probability theory, the probability of finding x within the range a_x to $a_x + dx$ and y within the range a_y to $a_y + dy$ is just the product of the individual probabilities for x and y

$$\left(e^{-x^2}dx\right)\left(e^{-y^2}dy\right) = \text{probability of finding } a_x < x < a_x + dx \text{ and } a_y < y < a_y + dy \quad (2.2)$$

This probability, which can be rewritten $\exp(-(x^2 + y^2)) \times dx \times dy$, has the intuitive interpretation as the volume of a rectangular prism placed on the xy -plane. The total probability of finding any value of x and y is just the sum of these volumes for the entire xy -plane, which corresponds with the total volume V of the surface $z = \exp(-(x^2 + y^2))$ over the entire xy -plane, shown in Figure ??

However, because x and y are independent variables, basic probability theory states that the total probability of finding x and y anywhere on the xy -plane equal to the volume V is just equal the product of the probabilities of finding x between $-\infty$ and ∞ and y between the same interval, so

$$\left(\int_{-\infty}^{\infty} e^{-x^2}dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2}dy\right) = V \quad (2.3)$$

However, the left-hand side of Equation (2.3) is just the product of two simple Gaussian integrals I ! Then,

$$I^2 = V. \quad (2.4)$$

With Equation 2.4, the Gaussian integral can be solved by finding the volume under the surface $z = \exp(-(x^2 + y^2))$ above the xy -plane. Fortunately, this volume is rotationally symmetric about the z -axis. This means the shell method of integration, shown schematically in Figure ??, can be used. Splitting the volume into shells centered on the z -axis with radius r , the height z of each shell is:

$$z = e^{-(x^2+y^2)} = e^{-r^2}, \quad (2.5)$$

where the Pythagorean is used to make the substitution $r^2 = x^2 + y^2$. The thickness of each shell is dr , and the circumference is $2\pi r$. Then, the shell volume dV is the product of the height, thickness, and circumference, or

$$dV = 2\pi r e^{-r^2} dr. \quad (2.6)$$

The total volume under the surface is then

$$V = \int_0^{\infty} 2\pi r e^{-r^2} dr, \quad (2.7)$$

where the limits of integration from 0 to ∞ correspond to shells with radii from 0 to ∞ spanning the entire xy -plane. Because polar coordinates are used, there is an additional r factor in the integral, which allows it to be solved with u -substitution! Making the substitution $u = r^2$ gives

$$\begin{aligned} V &= \int_0^{\infty} \frac{2\pi r}{2r} e^{-u} du \\ &= \int_0^{\infty} \pi e^{-u} du \\ &= -\pi e^{-u} \Big|_0^{\infty} \\ &= \pi \end{aligned}$$

Then, substituting V into Equation (2.4) solves the Gaussian integral

$$\boxed{I = \sqrt{\pi}} \quad (2.8)$$

Example 1: Show that the general Gaussian integral is

$$I = \int_{-\infty}^{\infty} Ae^{-ax^2} dx = A\sqrt{\frac{\pi}{a}}. \quad (2.9)$$

Solution 1: Again applying the polar substitution trick, the integral I boils down to finding a volume under the surface

$$I^2 = A^2 \int_0^{\infty} 2\pi r e^{-ar^2} dr, \quad (2.10)$$

where the two factors of the constant A have been pulled out of the integral. Since there is an additional scaling factor a in the exponential, the u -substitution $u = ar^2$ is required. Then,

$$\begin{aligned} I^2 &= A^2 \int_0^{\infty} \frac{2\pi r}{2ar} e^{-u} du \\ &= A^2 \int_0^{\infty} \frac{\pi}{a} e^{-u} du \\ &= A^2 \left(-\frac{\pi}{a} e^{-u} \Big|_0^{\infty} \right) \\ &= A^2 \left(\frac{\pi}{a} \right), \end{aligned}$$

which means $\boxed{I = A\sqrt{\pi/a}}$.

Example 2: In thermodynamics, the Maxwell–Boltzmann velocity distribution describes the distribution of the velocities of molecules in an ideal gas. The PDF for the x -component of velocity v_x is given by:

$$f(v_x) = Ae^{-mv_x^2/2kT}, \quad (2.11)$$

where A is some scaling constant and the values of m, k, T are given for an ideal gas. The probability of finding any value of v_x from $-\infty$ to ∞ is just 1, since a molecule must have an x -component of velocity in that range. Therefore,

$$1 = \int_{-\infty}^{\infty} f(v_x) dv_x. \quad (2.12)$$

Using Equation (2.12), find the expression for the scaling constant A .

Solution 2: $f(v_x)$ is a Gaussian function with $a = m/2kT$. Applying the polar coordinate trick again (or just directly using the result from Example 2) gives

$$1 = A\sqrt{\frac{\pi}{m/2kT}}, \quad (2.13)$$

or

$$A = \sqrt{\frac{m}{2\pi kT}}. \quad (2.14)$$

3 Motivation: Expectation Values

Suppose you know the PDF $f(x)$ for some quantity x . One thing you might be interested in is the *average* or expectation value for x , denoted with the bracket notation $\langle x \rangle$, in some interval (a, b) of the distribution. For a non-uniform $f(x)$, some values of x will appear more often in the distribution than others, so the average would be a weighted sum of x multiplied by the probability of finding x at that value

$$\langle x \rangle = \sum_{x_i} x_i \cdot f(x_i) dx, \quad (3.1)$$

where the summation is over all points $x_i \in (a, b)$. For a continuous distribution of x , Equation (3.1) can be rewritten with an integral

$$\langle x \rangle = \int_a^b x f(x) dx. \quad (3.2)$$

Therefore, to find the expectation value for x or *any polynomial* for x in a Gaussian distribution in the interval $(-\infty, \infty)$, integrals of the form

$$\langle x^n \rangle = A \int_{-\infty}^{\infty} x^n e^{-ax^2} dx \quad (3.3)$$

must be evaluated. Note that the expectation value of any polynomial of x in a Gaussian distribution is just a sum of separate x^n Gaussian integrals, since the integration operation is linear.

4 Related Gaussian Integrals

Consider the simple case where $n = 1$ and $A = a = 1$. The related Gaussian integral denoted I_1 is:

$$I_1 = \int_0^{\infty} x e^{-x^2} dx \quad (4.1)$$

Note that the lower limit of integration has been set to 0 and the upper limit is the usual ∞ . Since $x \exp(x^2)$ is an odd function, the integral from $(-\infty, \infty)$ is trivially zero and not particularly interesting.

Example 3: Show that $I_1 = 1/2$ in Equation (4.1).

Solution 3: I_1 can be easily evaluated using u -substitution.

$$\begin{aligned} I_1 &= \int_0^\infty x e^{-x^2} dx \\ &= \int_0^\infty e^{-x^2} \frac{1}{2} dx^2 \\ &= -\frac{1}{2} e^{-x^2} \Big|_0^\infty \\ &= \frac{1}{2} \end{aligned}$$

So $I_1 = 1/2$ as desired. Note that usual u -substitution has been written as

$$x dx = \frac{1}{2} dx^2. \quad (4.2)$$

This notation is just chosen for convenience and holds the same meaning as a $u = x^2$ substitution.

Now consider the integral I_2

$$I_2 = \int_{-\infty}^\infty x^2 e^{-x^2} dx. \quad (4.3)$$

Unfortunately, direct u -substitution with $u = x^2$ will not work for Equation (4.3). Attempting u -substitution with x^2 as in Solution ?? gives

$$I_2 = \int_{-\infty}^\infty x e^{-x^2} \frac{1}{2} dx^2. \quad (4.4)$$

The problem is that the integrand is $x \exp -x^2$, which contains an extra x factor. Fortunately, the standard way to evaluate this integral is to separate the product of x and e^{-x^2} using integration by parts to get

$$I_2 = -\frac{x e^{-x^2}}{2} \Big|_{-\infty}^\infty + \int_{-\infty}^\infty \frac{e^{-x^2}}{2} dx. \quad (4.5)$$

The second term in Equation (4.5) is half of the simple Gaussian integral, giving $(\sqrt{\pi/a})/2$. Using L'Hôpital's rule, it can be shown that the first term is zero. Therefore,

$$I_2 = \frac{\sqrt{\pi}}{2}. \quad (4.6)$$

The process of performing u -substitution with integration parts can be repeated to find the expectation values of successively higher powers of x^n in a Gaussian distribution.

Example 4: Evaluate the integral

$$I_4 = \int_{-\infty}^\infty x^4 e^{-x^2} dx. \quad (4.7)$$

Then, evaluate the general integral I_{2n} such that $n \in \mathbb{Z}$ is an integer

$$I_{2n} = \int_{-\infty}^{\infty} x^n e^{-x^2} dx. \quad (4.8)$$

Solution 4: To evaluate I_4 , first perform an x^2 substitution to give

$$I_4 = \int_{-\infty}^{\infty} x^3 e^{-x^2} \frac{1}{2} dx^2 \quad (4.9)$$

Then, integration-by-parts gives

$$I_4 = -\frac{x^3 e^{-x^2}}{2} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{3x^2 e^{-x^2}}{2} dx \quad (4.10)$$

Using L'Hôpital's rule, the first boundary term goes to zero. The second term is just $3/2 \cdot I_2$, so

$$I_4 = \frac{3\sqrt{\pi}}{4} \quad (4.11)$$

This procedure can be generalized to the integral I_{2n} with an even power of x in the Gaussian integral. All of the boundary terms from integration by parts will be in the form of some power of x multiplied with $\exp(-x^2)$ evaluated at $-\infty$ to ∞ , which will go to zero by L'Hôpital's rule:

$$\text{boundary term} \propto x^m e^{-x^2} \Big|_{-\infty}^{\infty} = 0. \quad (4.12)$$

Every time the substitution and integration by parts process is performed, the power of x^{2m} in the integral decreases by two, always remaining even until it reaches x^0 and the integral is reduced to a simple Gaussian integral. The integral term from integration by parts will acquire a factor of $1/2$ every time substitution is performed for each even power of x^{2m} , contributing a factor of

$$\text{substitution factor} = \frac{1}{2^n}. \quad (4.13)$$

For example, I_4 has $n = 2$, so Equation (4.13) accounts for the factor of $1/4$ found in expression for I_4 in Equation (4.11).

On the other hand, an odd factor $2m - 1$ that is one below the power x^{2m} will be multiplied every time integration by parts is performed, up until x^2 . For example, the factor of 3 in the expression for I_4 in Equation (4.11) is account for by the product of 3×1 , since 3 and 1 are both one less than the x^4 and x^2 . This product of odd powers can be expressed for any general I_{2n} as

$$\text{integration by parts factor} = \frac{(2n - 1)!}{2^{n-1}(n - 1)!}. \quad (4.14)$$

The $(2n - 1)!$ in the numerator accounts for all of the odd factors that are one less than $2n$, but also includes the even factors. There are $n - 1$ of these even factors starting from 2, so they can be removed by dividing by $(n - 1)!$ scaled by factors of two with 2^{n-1} .

Combining the factors Equation (4.13) and Equation (4.14) with the value of the standard Gaussian integral at the end of the substitution and integration by parts process gives

$$I_{2n} = \frac{(2n - 1)!}{2^{2n-1}(n - 1)!} \sqrt{\pi}. \quad (4.15)$$

Trigonometric Substitution

Leon Lian

August 2023

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1 Introduction

We will explore fundamental integration rules used in calculus, and cover essential concepts such as constant rule, exponential rule, power rule, and trigonometric rules. Each rule is illustrated with practical examples, helping readers gain confidence in integrating various functions. This document serves as a valuable reference for understanding and applying integration techniques.

2 When to Use Trig Substitution

Trig substitution is particularly useful when dealing with integrals containing the following types of expressions:

- Quadratic expressions under a square root, e.g., $\sqrt{a^2 - x^2}$, $\sqrt{x^2 - a^2}$, $\sqrt{x^2 + a^2}$
- Quadratic expressions with the form $a^2 - x^2$, $a^2 + x^2$, $x^2 - a^2$
- Rational expressions with quadratic denominators

3 Common Trigonometric Substitutions

The choice of trigonometric substitution depends on the form of the integral. Here are the common substitutions:

- For expressions of the form $\sqrt{a^2 - x^2}$, use $x = a \sin \theta$
- For expressions of the form $\sqrt{x^2 - a^2}$, use $x = a \cos \theta$
- For expressions of the form $\sqrt{x^2 + a^2}$, use $x = a \tan \theta$
- For expressions of the form $a^2 - x^2$, use $x = a \sin \theta$
- For expressions of the form $a^2 + x^2$, use $x = a \tan \theta$
- For expressions of the form $x^2 - a^2$, use $x = a \sec \theta$
- For rational expressions with quadratic denominators, use appropriate trigonometric substitutions based on the form.

4 Examples

Let's see some examples of how to use trig substitution to solve integrals:

4.1 Example 1

Example 1: Evaluate the integral: $\int \frac{1}{\sqrt{4-x^2}} dx$

Solution: Let $x = 2 \sin \theta$, then $dx = 2 \cos \theta d\theta$.

$$\begin{aligned}
 \int \frac{1}{\sqrt{4-x^2}} dx &= \int \frac{1}{\sqrt{4-(2 \sin \theta)^2}} \cdot 2 \cos \theta d\theta \\
 &= \int \frac{1}{\sqrt{4-4 \sin^2 \theta}} \cdot 2 \cos \theta d\theta \\
 &= \int \frac{1}{\sqrt{4 \cos^2 \theta}} \cdot 2 \cos \theta d\theta \\
 &= \int \frac{1}{2} d\theta \\
 &= \frac{\theta}{2} + C \\
 &= \frac{\arcsin\left(\frac{x}{2}\right)}{2} + C
 \end{aligned}$$

4.2 Example 2

Example 2: Evaluate the integral: $\int \frac{1}{\sqrt{x^2-1}} dx$

Solution: Let $x = \sec \theta$, then $dx = \sec \theta \tan \theta d\theta$.

$$\begin{aligned} \int \frac{1}{\sqrt{x^2-1}} dx &= \int \frac{1}{\sqrt{\sec^2 \theta - 1}} \cdot \sec \theta \tan \theta d\theta \\ &= \int \frac{1}{\sqrt{\tan^2 \theta}} \cdot \sec \theta \tan \theta d\theta \\ &= \int \frac{1}{|\tan \theta|} \cdot \sec \theta \tan \theta d\theta \\ &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C \\ &= \ln |x + \sqrt{x^2-1}| + C \end{aligned}$$

4.3 Example 3

Example 3: Evaluate the integral: $\int \frac{1}{x^2\sqrt{1-x^2}} dx$

Solution: Let $x = \sin \theta$, then $dx = \cos \theta d\theta$.

$$\begin{aligned} \int \frac{1}{x^2\sqrt{1-x^2}} dx &= \int \frac{1}{(\sin \theta)^2 \sqrt{1-(\sin \theta)^2}} \cdot \cos \theta d\theta \\ &= \int \frac{1}{\sin^2 \theta \cos \theta} d\theta \\ &= \int \frac{1}{\sin^2 \theta} \cdot \frac{1}{\cos \theta} d\theta \\ &= \int \csc^2 \theta d\theta \\ &= -\cot \theta + C \\ &= -\cot(\sin^{-1} x) + C \end{aligned}$$

4.4 Example 4

Example 4: Evaluate the integral: $\int \frac{1}{\sqrt{x^2+9}} dx$

Solution: Let $x = 3 \tan \theta$, then $dx = 3 \sec^2 \theta d\theta$.

$$\begin{aligned}\int \frac{1}{\sqrt{x^2 + 9}} dx &= \int \frac{1}{\sqrt{(3 \tan \theta)^2 + 9}} \cdot 3 \sec^2 \theta d\theta \\&= \int \frac{1}{\sqrt{9 \tan^2 \theta + 9}} \cdot 3 \sec^2 \theta d\theta \\&= \int \frac{1}{\sqrt{9(\tan^2 \theta + 1)}} \cdot 3 \sec^2 \theta d\theta \\&= \int \frac{1}{\sqrt{9 \sec^2 \theta}} \cdot 3 \sec^2 \theta d\theta \\&= \int 1 d\theta \\&= \theta + C \\&= \arctan\left(\frac{x}{3}\right) + C\end{aligned}$$

Integration by Parts

Nijat Abbasov

August 2023

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1 Introduction

Sometimes we need to find integrals of functions which are products of two different functions. For example, take your time and think how one can integrate $\int x \cos x dx$. Or, try solving $\int x \sec x^2$. As you can see, these are integrals of product of two completely different functions, being x and $\cos x$, x and $\sec x^2$. In this chapter, you will learn how to integrate these type of integrals.

2 Integration by parts

When we need to find integral of product of two functions, let's say $f(x)$ and $g(x)$: $f(x)g(x)dx$, we can substitute $f(x)$ with u and $g(x)dx$ with dv . Then, by utilizing the formula for the derivative of a product written in the differential notation

$$d(uv) = u dv + v du,$$

$$u dv = d(uv) - v du$$

$$\int u dv = \int d(uv) - \int v du$$

$$\int u dv = uv - \int v du$$

we get an equation named **Integration by Parts** which can help us solve integrals.

It comes to a rescue when we need integrate the product of two different functions if calculating $\int v du$ is easier than calculating $\int u dv$. For example, let's look at the following problem:

Example 1: Solve $\int x \ln(x) dx$

Solution: $f(x) = \ln(x)$ $g(x)dx = x dx$ $u = \ln(x)$ $dv = x dx$ $du = \frac{1}{x} dx$ and $v = \frac{x^2}{2}$.
Then, we can say that:

$$\int x \ln(x) dx = \frac{x^2}{2} \ln(x) - \int \frac{x}{2} dx = \frac{x^2}{2} (\ln(x) - \frac{1}{2})$$

The tricky part in applying integration by parts is how to select u and dv so that finding the value will be easy. When $u dv$ is long, this is especially tricky as finding the proper u and dv by trial and error takes a lot of time. But with lots of practice, one can find proper u and dv easily and faster. To select proper u and dv one should keep in mind before applying the rule that we need to be able to calculate the integral of dv and we need to be able to integrate $v du$. If it is known that one of this will fail, we can select another values for u and dv and avoid wasting time on u and dv which will not give a result. It is also possible that the integral maybe solved more easily or faster by u substitution. So, usually testing integration methods which would likely give a result before doing integration by parts is advised. Sometimes, one can apply integration by parts several times in succession. For example, let's look at the following problem.

Example 2: Find $\int x^n e^x dx$: $u = x^n$, $dv = e^x dx$. $du = nx^{n-1}$, $v = e^x$. Then, we find that:

$$\int x^n e^x dx = x^n e^x - n \int e^x x^{n-1}.$$

Now, we can see that x^n became x^{n-1} which means that if we continue applying integration by parts rule, the term x^n will be x^0 eventually. So, we can write the equation above as:

$$x^n e^x dx = x^n e^x - nx^{n-1} e^{n-1} + n(n-1)x^{n-2} e^{n-2} - n(n-1)(n-2)x^{n-3} e^{n-3} \dots$$

Note that, we continue this, till the power of x and e goes to zero.

Example 3: Another cool example is the reduction of $A_n = \int \sin^n x dx$: $u = \sin^{n-1} x$
 $dv = \sin x dx$ $du = (n-1)\sin^{n-2} x \cos x dx$ $v = -\cos x$

Therefore, $A_n = -\sin^{n-1} x \cos x + (n-1) \int \cos x^2 \sin^{n-2} dx$. Now, substituting $\cos x^2$ with $1 - \cos x^2$, we get:

$$A_n = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx - (n-1) \int \sin^n x dx$$

$$A_n = -\sin^{n-1} x \cos x + (n-1)A_{n-2} - (n-1)A_n$$

$$nA_n = -\sin^{n-1} x \cos x + (n-1)A_{n-2}$$

Here, you can see that index of A became smaller, which means if we continue, we can lower the index till it becomes zero or one. If A is odd, then we will use $A_1 = -\cos x$. Otherwise, we will use $A_0 = x$. We could solve the problem by utilizing half angle formulas, but when value of n is big, this becomes cumbersome.

3 Problems

1. Prove:

(a)

$$\int \ln(x)^n dx = x \ln(x)^n - n \int \ln(x)^{n-1} dx$$

(b)

$$\int \ln(x)^5 dx$$

2. Integrate

$$\int \frac{x^3}{(x^2 + 5)^2} dx$$

3. Integrate

$$x^7 \sqrt{5 + 3x^4} dx$$

4 Solutions

1. (a)

$$\int \ln(x)^n dx = \int u dv$$

$$u = \ln(x)^n$$

$$dv = dx$$

$$du = \frac{n}{x} \ln(x)^{n-1} dx$$

$$v = x$$

So,

$$\int u dv = uv - \int v du = x \ln(x)^n - n \int \ln(x)^{n-1} dx$$

(b) To do this, we just need to put the value of n in the formula above:

$$\int \ln(x)^5 dx = x \ln(x)^5 - 5 \int \ln(x)^4 dx$$

2. Let's separate factors x^2 and $\frac{x}{(x^2+5)^2} dx$. Then, we can name

$$u = x^2$$

$$dv = \frac{x}{(x^2+5)^2} dx$$

$$du = 2x dx$$

$$v = \left(-\frac{1}{2}\right)(x^2+5)^{-1} = \frac{-1}{2(x^2+5)}.$$

Therefore,

$$\int \frac{x^3}{(x+5)^2} = -\frac{x^2}{2(x^2+5)} + \int \frac{x dx}{(x^2+5)}$$

By using "u" substitution,

$$\int \frac{x^3}{(x+5)^2} = -\frac{x^2}{2(x^2+5)} + \frac{1}{2} \ln|x^2+5| + C.$$

3.

$$x^7 \sqrt{5+3x^4} dx$$

$$u = x^4$$

$$dv = x^3 \sqrt{5+3x^4} dx$$

$$du = 4x^3$$

$$v = \frac{1}{12} \frac{(5+3x^4)^{1.5}}{3/2} = \frac{1}{18} (5+3x^4)^{1.5}$$

Therefore,

$$x^7 \sqrt{5 + 3x^4} dx = x^4 \frac{1}{18} (5 + 3x^4)^{1.5} - \int \frac{1}{18} (5 + 3x^4)^{1.5} (4x^3) dx$$

$$x^7 \sqrt{5 + 3x^4} dx = x^4 \frac{1}{18} (5 + 3x^4)^{1.5} - \frac{2}{9} \int (5 + 3x^4)^{1.5} (x^3) dx$$

We can solve this by using "u" substitution:

$$x^7 \sqrt{5 + 3x^4} dx = x^4 \frac{1}{18} (5 + 3x^4)^{1.5} - \frac{2}{9} \frac{1}{12} \frac{(5 + 3x^4)^{2.5}}{2.5} + C$$

$$x^7 \sqrt{5 + 3x^4} dx = x^4 \frac{1}{18} (5 + 3x^4)^{1.5} - \frac{1}{135} (5 + 3x^4)^{2.5} + C$$

Integral Averages

Logan Reich

August 2023

1 Introduction

How do you take the average of $0.5x^2 + 1$ from 3 to 8? Well, we can use something called integral averages!

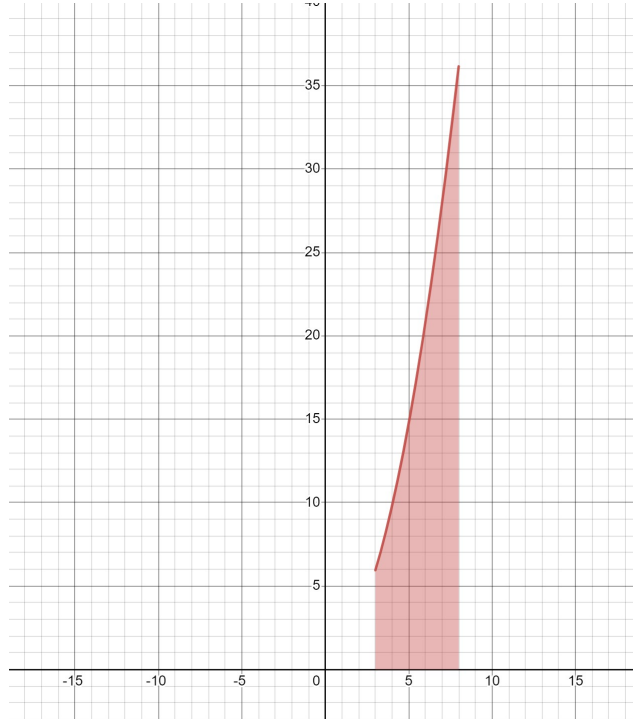


Figure 1: A depiction of the actual meaning of an integral as the area under a curve (the author, using Desmos)

An integral average is a method of taking the average of a function across a range by using integrals. It relies on the following formula:

$$average = \frac{\int_{lowerbound}^{upperbound} f(x) \, dx}{upperbound - lowerbound}$$

This simply finds the area under the relevant portion of the curve and divides it by the length - to find the height it would be if the shape were a rectangle, which is the average height and thus

the average value. Applying the formula to this problem, we get:

$$average = \frac{\int_3^8 0.5x^2 + 1 \, dx}{8 - 3} = \frac{126.25}{5} = 25.25$$

And there's our answer! Here are a few practice examples: Take the average of the following from -4 to 4:

$$8x^3 + 4x + 1$$

Ans: 1

$$x^2 + 2x + 3$$

Ans: 25/3

$$9x^3 + x + 7$$

Ans: 7

$$4x + 1024$$

Ans: 1024

$$8x^4$$

Ans: 409.6

$$1834567892345679x + 45678183$$

Ans: 45678183

$$1$$

Ans: 1

Substitutions

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1 Introduction

Among the multitude of mathematical tools to conquer integral computations, the u-substitution is one of the most fundamental, as it helps simplify otherwise ugly looking integrals. Let's get familiar with them in this handout!

2 Definition and Examples

2.1 Why u-sub works

Definition: This integration technique is basically applying **reverse chain rule** by the Fundamental Theorem of Calculus. It aims to write integrals into the form

$$\int f(g(x))g'(x)dx$$

which, when using a substitution of $u = g(x)$ gives:

$$\int f(u)du$$

because $g'(u) = \frac{du}{dx}$.

2.2 Steps to perform a u-sub

There are a few steps to performing a u-substitution. As an example, let's say we wanted to evaluate

$$\int_3^5 \sin(1 + 69x)dx$$

Here's how you could do it using u-sub.

1. **Choose a simplifying transformation:** Here, notice that if we could apply the power rule, that would be wonderful. Hence we choose the substitution

$$u = 1 + 69x$$

2. **Determine the corresponding differential:** We want to change everything from the x world into the u world: although we changed the part inside the parentheses, we have yet to change the differential dx to du . To find that, we can compute the ratio $\frac{du}{dx}$ with differentiation:

$$\begin{aligned}\frac{du}{dx} &= \frac{d}{dx}(1 + 69x) = 69 \\ du &= 69dx\end{aligned}$$

3. **Change the bounds of integration:** This part only applies if you have definite integrals. For $x = 3$, $u = 1 + 69(3) = 208$. Similarly for $x = 5$, $u = 346$.

4. **Assembling the integral:** Putting all three steps from above together, we get:

$$\int_{x=3}^{x=5} \sin(1+69x)dx = \int_{u=208}^{u=346} \sin(u) \cdot \left(\frac{1}{69}du\right)$$

From here on, it is straightforward to perform the integral because we know the integral of $\sin(x)$, which one can obtain from the Fundamental Theorem of Calculus, or by looking up an integral table/formula sheet.

The reason we have to perform all the steps above is because u-substitution is basically the same thing as changing a graph's axes - if one pictures integrals as areas under curves, then u-substitution is mapping one x-axis into another u-axis. Not only does the scales on the axes change (which implies that dx changes into du), but also the bounds of integration too.

Note that you don't always have to perform substitutions of u as a function of x . You could also perform substitutions where x is a function of u , as we will see below.

2.3 Trigonometric Substitutions

As you continue on your calculus journey, you will encounter more and more substitutions. One particular family of substitutions are **trigonometric substitutions**. The reason they are so useful and even deserve to be named "a family of substitutions" is because of the trigonometric identities, which can help simplify an integral greatly.

Example 1: Let's consider the integral:

$$\int \frac{1}{(1+x^2)^{3/2}} dx$$

At first glance, this seems hard because you can't do reverse-power-rule on this. However, there is a beautiful coincidence:

$$\begin{aligned} 1 + \tan^2 x &= \sec^2 x \\ \frac{d}{dx} \tan x &= \sec^2 x \end{aligned}$$

The derivative of $\tan x$ is exactly the same as 1 plus itself squared! This motivates us to perform the substitution $x = \tan u$. This implies that:

$$dx = \sec^2 u du$$

Plugging both of them into the original integral:

$$\int \frac{1}{(1+x^2)^{3/2}} dx = \int \frac{1}{(1+\tan^2 u)^{3/2}} \sec^2 u du = \int \frac{1}{\sec^3 u} \sec^2 u du = \int \cos u du$$

The original integral then evaluates to

$$\int \frac{1}{(1+x^2)^{3/2}} dx = \sin u + C = \sin(\arctan x) + C$$

which one can further simplify using trigonometry, but that would be outside the focus of this handout.

The thing about trigonometric substitutions is that whenever you see a certain "form" of function in the integrand, then one should use a certain substitution. The problem then transforms from a polynomial type function into a trigonometric type function.

Here's a list of useful **trigonometric substitutions**:

Form	Substitution	Reason
$\sqrt{a^2 + x^2}$	$u = a \tan x$	$1 + \tan^2 x = \sec^2 x = (\tan x)'$
$\sqrt{a^2 - x^2}$	$u = a \sin x$	$1 - \sin^2 x = \cos^2 x = ((\sin x)')^2$
$\sqrt{x^2 - a^2}$	$u = a \sec x$	$\sec^2 x - 1 = \tan^2 x = \left(\frac{(\sec x)'}{\sec x}\right)^2$

where $(f(x))' = \frac{d}{dx}f(x)$ and $(\sec x)' = \sec x \tan x$

2.4 A Note on Substitutions

As seen from above, the list of substitutions is extensive - in there are many types of substitutions out there too, including Weierstrass substitutions, Euler substitutions, and many other tricks that one will discover along their path on the journey through calculus.

The key to substitutions is not memorising all of them, but rather, choosing the correct one which simplifies the problem at hand.

3 Problems & Practices

3.1 Problem 1

Evaluate the integral

$$\int 3x^2(x^3 + 1)^6 dx$$

3.2 Problem 2

Evaluate the integral

$$\int \frac{(\ln x)^2}{x} dx$$

3.3 Problem 3

Evaluate the integral

$$\int \frac{1}{\sqrt{a^2 - b^2 x^2}} dx$$

3.4 Problem 4

Prove the King Property:

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

3.5 Problem 5

Evaluate the integral

$$\int_0^\pi \frac{x \sin x}{(1 + \cos^2 x)^2} dx$$

Hint: use the result from Problem 3, and think about the periodicity of the sinusoidal functions.

3.6 Problem 6

Evaluate the integral

$$\int \frac{1}{x^2(x^4 + 1)^{3/4}} dx$$

3.7 Problem 7

Evaluate the integral

$$\int \frac{x^2 - 2}{(x^4 + 5x^2 + 4) \arctan\left(\frac{x^2+2}{x}\right)} dx$$

3.8 Problem 8

Evaluate the integral

$$\int_0^{\pi/2} \ln(\sin x) dx$$

Hint: use the result from Problem 3. You might use the identity $\sin 2x = 2 \sin x \cos x$

4 Solutions**4.1 Problem 1**

$$\frac{(x^3 + 1)^7}{7} + C$$

4.2 Problem 2

$$\frac{(\ln x)^3}{3} + C$$

4.3 Problem 3

$$\frac{1}{b} \arcsin \frac{b}{a} x + C$$

4.4 Problem 4

Perform the substitution $x = a + b - u$ to get

$$\int_b^a f(a + b - u)(-du) = \int_a^b f(a + b - u)du$$

Then notice that the integral is definite, and the variable u is a dummy variable and can be named anything - by changing it to x we complete the proof of the King Property.

4.5 Problem 5

$$\frac{\pi^2}{4}$$

4.6 Problem 6

Perform the substitution $u = x^{-4}$ to get

$$\frac{-1}{4} \int \frac{du}{(1+u)^{3/4}}$$

which evaluates to

$$-(1+x^{-4})^{1/4} + C$$

4.7 Problem 7

$$\ln \left(\arctan \left(\frac{x^2 + 2}{x} \right) \right) + C$$

4.8 Problem 8

$$\frac{-\pi \ln 2}{2}$$

5 Conclusion

Congratulations on making it through this handout! The problems were tough, so give yourself a pat on the back if you managed to do them all - you can now compute many different types of integrals!

Integration Rules

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1 Introduction

We will explore the fundamental integration rules used in calculus, and cover essential concepts such as constant rule, exponential rule, power rule, and trigonometric rules. Each rule is illustrated with practical examples, helping readers gain confidence in integrating various functions. This document serves as a valuable reference for understanding and applying integration techniques.

2 Integration Rules

2.1 Constant Rule

The constant rule states that the integral of a constant a with respect to x is the constant times x plus the constant of integration C :

$$\int a \, dx = ax + C$$

Example

Evaluate the integral: $\int 4 \, dx$

Solution

$$\int 4 \, dx = 4x + C$$

2.2 Exponential Rule

The exponential rule states that the integral of e^x with respect to x is e^x plus the constant of integration C :

$$\int e^x \, dx = e^x + C$$

2.3 Power Rule

The power rule states that the integral of x^n with respect to x (where n is any real number except -1) is $\frac{x^{n+1}}{n+1}$ plus the constant of integration C :

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \quad \text{where } n \neq -1$$

Example

Evaluate the integral: $\int x^2 \, dx$

Solution

$$\int x^2 \, dx = \frac{x^3}{3} + C$$

2.4 Multiplication by Constant

The multiplication by constant rule states that if we have a constant c multiplied by a function $f(x)$, then the integral of this product is equal to c times the integral of $f(x)$ with respect to x , plus the constant of integration C :

$$\int cf(x) \, dx = c \int f(x) \, dx$$

Example

Evaluate the integral: $\int 5x^2 \, dx$

Solution

$$\int 5x^2 \, dx = 5 \int x^2 \, dx = 5 \cdot \frac{x^3}{3} + C = \frac{5x^3}{3} + C$$

2.5 Sum Rule

The sum rule states that the integral of the sum of two functions $f(x)$ and $g(x)$ with respect to x is equal to the sum of their individual integrals, each with respect to x :

$$\int (f(x) + g(x)) \, dx = \int f(x) \, dx + \int g(x) \, dx$$

Example

Evaluate the integral: $\int (2x^2 + 3x) \, dx$

Solution

$$\int (2x^2 + 3x) \, dx = \int 2x^2 \, dx + \int 3x \, dx = \frac{2x^3}{3} + \frac{3x^2}{2} + C$$

2.6 Difference Rule

The difference rule states that the integral of the difference of two functions $f(x)$ and $g(x)$ with respect to x is equal to the difference of their individual integrals, each with respect to x :

$$\int (f(x) - g(x)) \, dx = \int f(x) \, dx - \int g(x) \, dx$$

Example

Evaluate the integral: $\int (5x^2 - 2x) \, dx$

Solution

$$\int (5x^2 - 2x) \, dx = \int 5x^2 \, dx - \int 2x \, dx = \frac{5x^3}{3} - x^2 + C$$

2.7 Reciprocal Rule

The reciprocal rule, also known as the "Natural Logarithm Rule," states that the integral of $\frac{1}{x}$ with respect to x is the natural logarithm of the absolute value of x plus the constant of integration C :

$$\int \frac{1}{x} dx = \ln |x| + C$$

Example

Evaluate the integral: $\int \frac{1}{2x} dx$

Solution

$$\int \frac{1}{2x} dx = \frac{1}{2} \int \frac{1}{x} dx = \frac{1}{2} \ln |x| + C$$

2.8 Trigonometric Rules (in radians)

The trigonometric rules involve integrals of trigonometric functions. Here are some important ones (where C is the constant of integration):

$$\int \sin(x) dx = -\cos(x) + C$$

$$\int \sec^2(x) dx = \tan(x) + C$$

$$\int \sec(x) \tan(x) dx = \sec(x) + C$$

$$\int \cos(x) dx = \sin(x) + C$$

$$\int \csc(x) dx = -\cot(x) + C$$

$$\int \csc(x) \cot(x) dx = -\csc(x) + C$$

2.9 Integrals of Inverse Trigonometric Functions

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x) + C$$

$$\int \left(-\frac{1}{\sqrt{1-x^2}} \right) dx = \arccos(x) + C$$

$$\int \frac{1}{1+x^2} dx = \arctan(x) + C$$

$$\int \left(-\frac{1}{|x|\sqrt{x^2-1}} \right) dx = \operatorname{arccsc}(x) + C$$

$$\int \frac{1}{|x|\sqrt{x^2-1}} dx = \operatorname{arcsec}(x) + C$$

$$\int \left(-\frac{1}{1+x^2} \right) dx = \operatorname{arccot}(x) + C$$

3 Examples

3.1 Example 1

Evaluate the integral: $\int (4x^6 - 2x^3 + 7x - 4) dx$

Solution: To integrate each term, we use the power rule for integration.

$$\int (4x^6 - 2x^3 + 7x - 4) dx = \frac{4}{7}x^7 - \frac{1}{2}x^4 + \frac{7}{2}x^2 - 4x + C$$

3.2 Example 2

Evaluate the integral: $\int (z^7 - 48z^{11} - 5z^{16}) dz$

Solution: To integrate each term, we use the power rule for integration.

$$\int (z^7 - 48z^{11} - 5z^{16}) dz = \frac{1}{8}z^8 - 4z^{12} - \frac{5}{17}z^{17} + C$$

3.3 Example 3

Evaluate the integral: $\int (10t^{-3} + 12t^{-9} + 4t^3) dt$

Solution: To integrate each term, we use the power rule for integration.

$$\begin{aligned} \int (10t^{-3} + 12t^{-9} + 4t^3) dt &= -\frac{10}{2t^2} - \frac{12}{8t^8} + \frac{4}{4}t^4 + C \\ &= -\frac{5}{t^2} - \frac{3}{2t^8} + t^4 + C \end{aligned}$$

3.4 Example 4

Evaluate the integral: $\int (t^2 - 1)(4 + 3t) dt$

Solution: We use the distributive property and then apply the power rule for integration.

$$\begin{aligned} \int (t^2 - 1)(4 + 3t) dt &= \int (3t^3 + 4t^2 - 3t - 4) dt \\ &= \frac{3}{4}t^4 + \frac{4}{3}t^3 - \frac{3}{2}t^2 - 4t + C \end{aligned}$$

3.5 Example 5

Evaluate the integral: $\int (\sin(x) + 10(\csc(x))^2) dx$

Solution: For the first term, we use the integral of $\sin(x)$, which is $-\cos(x)$. For the second term, we use the integral of $\csc(x)^2$, which is $-\cot(x)$.

$$\int (\sin(x) + 10(\csc(x))^2) dx = -\cos(x) + 10(-\cot(x)) + C$$

3.6 Example 6

Evaluate the integral: $\int (2\cos(w) - \sec(w)\tan(w)) dw$

Solution: For the first term, we use the integral of $\cos(w)$, which is $\sin(w)$. For the second term, we use the integral of $\sec(w)\tan(w)$, which is $\sec(w)$.

$$\int (2\cos(w) - \sec(w)\tan(w)) dw = 2\sin(w) - \sec(w) + C$$

3.7 Example 7

Evaluate the integral: $\int \left(t^3 - \frac{e^{-t}-4}{e^{-t}}\right) dw$

Solution: First, simplify the expression inside the integral.

$$\begin{aligned} \int \left(t^3 - \frac{e^{-t}-4}{e^{-t}}\right) dw &= \int \left(t^3 - \frac{e^{-t}}{e^{-t}} + \frac{4}{e^{-t}}\right) dw \\ &= \int (t^3 - 1 + 4e^t) dw \end{aligned}$$

Now, integrate each term separately. The integral of t^3 is $\frac{t^4}{4}$. The integral of 1 with respect to t is t , and the integral of $4e^t$ with respect to t is $4e^t$.

$$\int (t^3 - 1 + 4e^t) dw = \frac{t^4}{4} - t + 4e^t + C$$

3.8 Example 8

Evaluate the integral: $\int \left(\frac{1}{1+x^2} + \frac{12}{\sqrt{1-x^2}}\right) dx$

Solution: Split the integral into two separate integrals and evaluate them separately.

$$\int \left(\frac{1}{1+x^2} + \frac{12}{\sqrt{1-x^2}}\right) dx = \int \frac{1}{1+x^2} dx + \int \frac{12}{\sqrt{1-x^2}} dx$$

For the first integral, we recognize it as the integral of $\arctan(x)$.

$$\int \frac{1}{1+x^2} dx = \arctan(x)$$

For the second integral, we recognize it as the integral of $\arcsin(x)$.

$$\int \frac{12}{\sqrt{1-x^2}} dx = 12 \arcsin(x)$$

Now, we combine the two results.

$$\int \left(\frac{1}{1+x^2} + \frac{12}{\sqrt{1-x^2}} \right) dx = \arctan(x) + 12 \arcsin(x) + C$$

3.9 Example 9

Evaluate the integral: $\int \left(\sqrt{x^7} - 7\sqrt[6]{x^5} + 17\sqrt[3]{x^{10}} \right) dx$

Solution: Get rid of the roots by making the exponents fractional.

$$\int \left(\sqrt{x^7} - 7\sqrt[6]{x^5} + 17\sqrt[3]{x^{10}} \right) dx = \int \left(x^{\frac{7}{2}} - 7x^{\frac{5}{6}} + 17x^{\frac{10}{3}} \right)$$

Apply the power rule for each term to integrate.

$$\int \left(x^{\frac{7}{2}} - 7x^{\frac{5}{6}} + 17x^{\frac{10}{3}} \right) dx = \frac{2}{9}x^{\frac{9}{2}} - \frac{42}{11}x^{\frac{11}{6}} + \frac{51}{13}x^{\frac{13}{3}} + C$$

Fundamental Theorem of Calculus

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1 Introduction

In the other handouts, you may have learned how to compute derivatives and interpret integrals. In this handout, we go through how one would actually compute these integrals mathematically, by treating the mathematical operation as reverse-differentiation.

2 Definition & Intuition

2.1 Formal Definition

In mathematics, many operations have inverses: addition has subtraction, multiplication has division, and the dual for calculus is that derivatives and integrals both undo each other. This is the basis for the **Fundamental Theorem of Calculus**.

There are two parts to the **Fundamental Theorem of Calculus (FToC)**.

Part 1: The derivative of an integral with an upper bound x is the integrand evaluated at that point x :

$$\frac{d}{dx} \int_a^x f(x) dx = f(x)$$

Part 2: The integral from endpoints a to b is the same as the indefinite integral evaluated at b minus the same thing evaluated at a :

$$\int_a^b f(x) dx = F(b) - F(a)$$

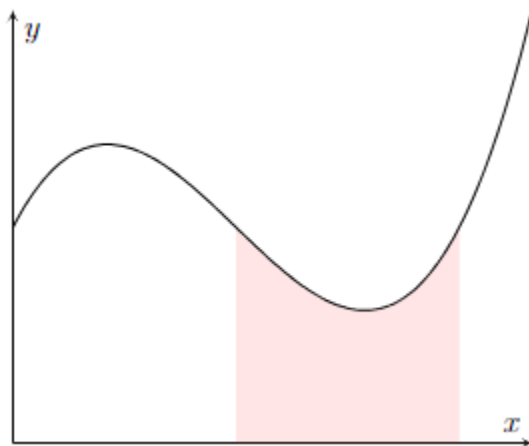
where $F(x) = \int f(x) dx + C$

2.2 Gaining Intuition

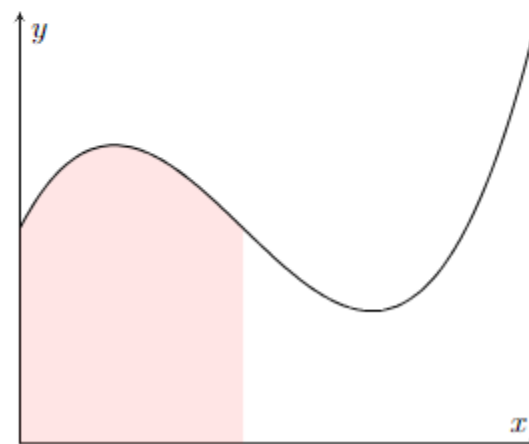
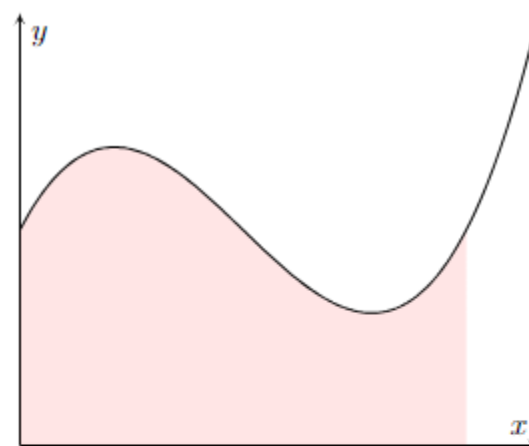
Now, why do these two parts work? Let's first talk about Part 2.

$$\int_a^b f(x) dx = F(b) - F(a)$$

In the "Intuition Behind Integrals" handout, we said that we could interpret integrals as areas under curves.



The area between this region can be computed by the difference of the following two areas:

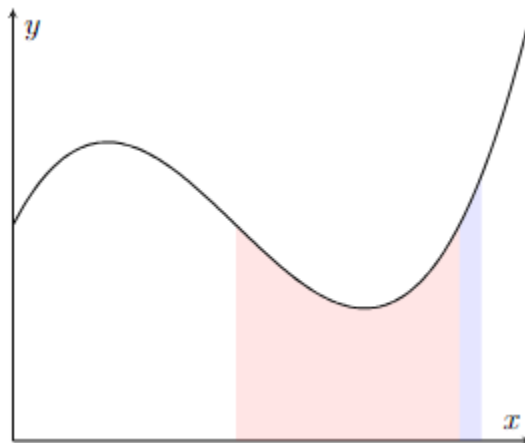


And both of these areas are represented by $F(b)$ and $F(a)$ respectively. Notice that we're only talking about the right endpoints here, and the left endpoint plays no role here, so it is free to be whatever number it wants to be.

Next, what about the first part of the FToC?

$$\frac{d}{dx} \int_a^x f(x)dx = f(x)$$

Referencing the "Intuition Behind Integrals" handout again, a derivative is just a fraction of infinitesimals. The denominator of the LHS is the familiar infinitesimal change dx , while the numerator is the infinitesimal change in the integral caused by the infinitesimal change in x . Here, x plays the right endpoint of the integral, and hence it's the infinitesimal change in area.



Here, the red region represents the existing area, while the blue region represents the infinitesimal change in the area (exaggerated so it is visible). If we take the blue area and divide it by dx , i.e. computing $\frac{d}{dx} \int_a^x f(x)dx$, then we would get $f(x)$ because the area of the blue area is just $f(x) \times dx$.

Another way to look at this statement is to notice that integrals are just fancy sums of infinitesimally small stuff. When we change x by dx , we're changing the integral by adding one more infinitesimal thing - and the thing we add is the integrand, $f(x)dx$. Hence

$$\frac{d \int_a^x f(x)dx}{dx} = \frac{f(x)dx}{dx} = f(x)$$

2.3 Why the $+C$?

In the text above, you might have noticed that we defined $F(x)$, the indefinite integral of $f(x)$ as

$$F(x) = \int f(x)dx + C$$

But it doesn't crop up in definite integrals - why do we even need that $+C$ term?

To explain this, we need to know that integrals are adding change, and they only add changes - they cannot tell you what they are adding to, which you have to add by yourself.

An analogy of this is a child playing with a tap which is filling a bucket with water below. Let's say the child keeps varying the flow rate at the tap over time, say, $\frac{dV}{dt} = f(t)$ as a function of time. To find the volume of water collected in the bucket, we integrate the function $f(t)$ over t . However, this does not tell us anything about the absolute amount of water in the bucket! Specifically, we do not know if there was already water in the bucket to start with - the integral only tells us how much the volume changed after some time.

Here, the analogy to the initial volume of water in the bucket is the integration constant $+C$.

3 Examples & Applications

3.1 Integrals as Reverse Derivatives

With the FToC defined in place, let's put both parts together!

From Part 2 of FToC, we know

$$\int_a^b f(x)dx = F(b) - F(a)$$

Putting this into Part 1 of FToC, we get

$$\frac{d}{dx} \int_a^x f(x)dx = f(x)$$

$$\frac{d}{dx}(F(x) - F(a)) = f(x)$$

$$\frac{d}{dx}F(x) = f(x)$$

since $F(a)$ is just a constant, whose derivative is just zero. Note that by definition $F(x) = \int f(x)dx + C$.

Now that we've proven that derivatives and integrals undo each other, we can compute integrals with this in mind.

3.2 Computing Integrals

To differentiate, it is straightforward and one simply follows the rules and pray for no mathematical mistakes. However, not all integrals can be easily computed. Some are so hard that mathematicians even name them "non-elementary" and call it a day! However, there exist many tools to help us compute integrals as you shall see in the other handouts.

In this handout, we shall use FToC to compute simple integrals.

Let's say you want to compute

$$\int_0^2 2x dx$$

Even if you have never taken an integral before, you probably did take a derivative before - and hopefully notice that the derivative of x^2 is precisely $2x$.

Then, by the FToC,

$$\int_0^2 2x dx = F(2) - F(0) = 4$$

where $F(x) = x^2$

This example is simple, but its principle is crucial and must be kept in the back of your head at all times.

Another example would be the integral

$$\int \sec^2(x) dx$$

If you didn't know about the FToC, you'd need to take some non-straightforward way to compute the integral - but, if you just realize that the derivative of $\tan(x)$ is precisely $\sec^2(x)$, then you can directly say

$$\int \sec^2(x) dx = \tan(x) + C$$

4 Problems & Practices

4.1 Problem 1

Problem 1: Use FToC to find the derivative of the functions

(a)

$$g(x) = \int_6^x \frac{1}{1+t^3} dt$$

(b)

$$h(t) = \int_9^{\sin(t)} \sqrt{x^2 + 4} dx$$

4.2 Problem 2

Problem 2:

(a) Take the derivative of the function

$$\ln(\tan x)$$

(b) Hence, find evaluate the indefinite integral

$$\int \frac{1}{\sin x} dx$$

4.3 Problem 3

Problem 3:

(a) Take the derivative of the function

$$f(x) = \ln \frac{x-a}{x+a}$$

(b) Hence, find evaluate the indefinite integral

$$\int \frac{1}{x^2 - a^2} dx$$

4.4 Problem 4

Problem 4: Evaluate the indefinite integral

$$\int \frac{\sin x \sec^2 x - \cos x \tan x}{\sin^2 x} dx$$

4.5 Problem 5

Problem 5: Evaluate the indefinite integral

$$\int \left(\frac{\sin^{-1} x}{x} + \frac{\ln x}{\sqrt{1-x^2}} \right) dx$$

5 Solutions

5.1 Problem 1

a)

$$g'(x) = \frac{1}{1+x^3}$$

b)

$$h'(t) = \sqrt{t^2 + 4} \cdot \cos(t)$$

5.2 Problem 2

a)

$$\frac{d}{dx} \ln \tan x = \frac{1}{\tan x} \sec^2 x = \frac{1}{\sin x \cos x} = \frac{2}{\sin 2x}$$

b)

$$\int \frac{1}{\sin x} dx = \ln \tan \frac{x}{2}$$

5.3 Problem 3

a)

$$\frac{df}{dx} = \frac{x+a}{x-a} \cdot \frac{(x+a) - (x-a)}{(x+a)^2} = \frac{2a}{x^2 - a^2}$$

b)

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \frac{x-a}{x+a}$$

5.4 Problem 4

$$\int \frac{\sin x \sec^2 x - \cos x \tan x}{\sin^2 x} = \frac{\tan x}{\sin x} = \frac{1}{\cos x}$$

5.5 Problem 5

$$\int \left(\frac{\sin^{-1} x}{x} + \frac{\ln x}{\sqrt{1-x^2}} \right) dx = \ln x \sin^{-1} x$$

6 Conclusion

Having the Fundamental Theorem of Calculus in the back of your head can help you evaluate some integrals instantly - congratulations on making it through this handout!

Intuition Behind Integrals

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1 Introduction

Integrals are often seen in higher physics, and might look daunting to the uninformed. However, they are a crucial in developing theories that accurately model the world - let's take a look at how to get familiar with them!

2 Definition & Intuition

When you first learn integration in high school, you might have merely been introduced to it as a mathematical operator - something that inputs a function, and spits out another function. However, that severely undermines the ideas that lie beneath it!

Take a look at the integration symbol:

$$\int$$

It looks like a weird "S", and the reason for that is because it stands for "sum".

Sometimes, you may see numbers on the top and bottom of the symbol:

$$\int_a^b$$

This means to add from the lower bound "a" to the upper bound "b". With bounds, the integral is called a **definite integral**, because it has definite endpoints. Without them, it would be called an **indefinite integral** for the same reason.

All of the time, you would see the term "dx" in the integral:

$$\int_a^b dx$$

The term dx is called a **differential element** - you might have encountered it when learning differentiation, written as

$$\frac{dy}{dx}$$

The idea is to interpret the derivative operator as a **fraction** (ignore the wailing mathematicians) - the reason to motivate this perspective can be seen from the definition of a derivative:

$$\frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

If you've seen this before, great! If not, even better! Because seeing derivatives as the limit of a fraction is a great way to gain intuition behind this mathematical machinery.

Going back to integrals, what would the dx in the integral mean? If we take dx to mean the same thing as in derivatives, it would mean that it is something that is very very small in the limit - which is precisely what it is! An integral is nothing but a summation of infinitely small things. To a sprinkle of mathematical rigor, the definition of a definite integral is

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n f(x_k) \Delta x$$

where $\Delta x = \frac{b-a}{n}$ and $x_k = a + k \cdot \Delta x$.

Now that might seem a lot to take in - where did the term $f(x)$ come from, and since when did infinity join the party?! Let's break down each part one by one.

The reason we put $f(x)$ inside the integral is because most of the time we want to add something that varies with x - if x denotes time, this might be something like velocity, for example, where the velocity might not be constant in time but we want to find where something is at the end of the interval. To quantify how velocity would vary with time, we write it as a function of time, specifically $f(x)$.

Now, on the right side of the definition, the word "lim" refers to a **limit**, where we take something to one extreme, in this case to infinity (because infinity is not a number, but a concept). In this case, we sum up infinitely many terms - but what do we sum up?

The summands are the function in question that we want to integrate $f(x)$, multiplied by a finite interval Δx . Again, think back to the kinematics example - if x were to denote time, and $f(x)$ to denote velocity, then we would be adding velocity $\times$ time, which is just displacement. Notice that we are evaluating the velocity at different times x_k , where x_k would slowly increment from $x_0 = a$, $x_1 = a + \Delta x$, $x_2 = a + 2\Delta x$, all the way up to $x_n = b$.

Now, one problem that might arise is: what about the velocities in between x_0 and x_1 ? Do we not account for them when computing the total displacement? Well, yes we would: when we take the limit to infinity, the intervals Δx would tend towards zero, hence there would ideally be no gaps between x_0 and x_1 in the limit, and all velocities would be accounted for.

You can play around with the idea over at this Desmos plot - do play around with the sliders and the function!

3 Examples & Applications

3.1 Integrals as Areas

Arguably the most popular application of integrals in math courses is to find areas under certain graphs - the reason integrals can directly correspond to areas is shown in the Desmos plot above: integrals help break down the graph into infinitely many small rectangular chunks, which when added together gives the area. One thing to note is that the areas are **signed**, that is, there are **positive and negative areas**, highlighted as red and green in the Desmos plot linked earlier.

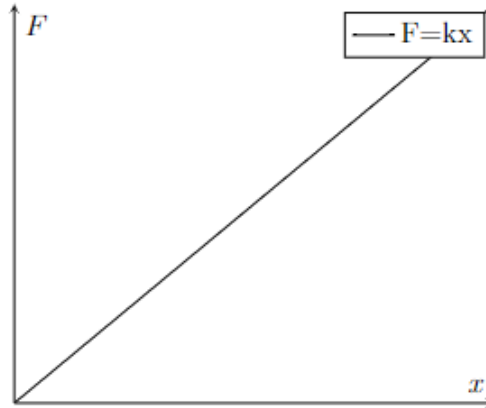
A picture is worth a thousand words, and graphs are undeniably useful in physics as they can give us lots of information, from the behaviour of the function, to its gradient, and the most relevant one: the area enclosed under it.

Areas under curves crop up a lot in physics: from simple scenarios like deriving the formula for the energy of a spring to more advanced cases like Maxwell's equal area construction in thermodynamics.

Let's look at the simpler example of the energy of a spring system. By Hooke's Law, we know that the force exerted by a spring is

$$\vec{F} = -k\vec{x}$$

with the negative sign indicating the opposite direction. Plotting its absolute values on a graph, we see a linear function:



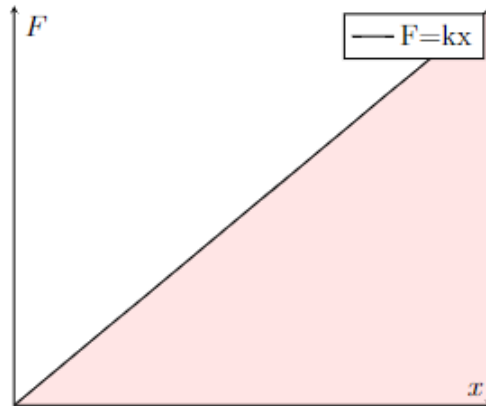
By definition, potential energy of a system at some state is the amount of energy put in to bring the system to that state. The infinitesimal work done when the spring is stretched by a miniscule length dx is

$$dU = -Fdx = kx dx$$

Hence, when the spring is stretched from $x = 0$ to $x = x$, the energy that needs to be put in is

$$U = \int_{x=0}^{x=x} dU = \int_0^x kx \cdot dx$$

Graphically, this corresponds to the area under the linear function.



This gives the result

$$U = \frac{1}{2}kx^2$$

which is the familiar expression for the energy of a spring.

3.2 Adding changes

Another way to view integrals is that they add together infinitely many tiny changes. Kinematics provide a great example of this: by definition, velocity is

$$v = \frac{dx}{dt}$$

(i.e. the infinitesimal change in displacement dx divided by an infinitesimal interval of time dt).

To find the total displacement, we can add together infinitely many tiny displacements with an integral:

$$x = \int dx$$

And, with the definition of velocity, we get

$$x = \int v \cdot dt$$

This is useful because, if we know how velocity varies with time, i.e. $v(t)$, then we can precisely know how much the object has moved.

3.3 Infinitesimal Superposition

One more important place where integrals pop up is **superposition**. (feel free to come back to this later when you've learned electrostatics)

Namely, in EM problems, we often have a charge distribution instead of point charges. As an example, let's consider an infinitely long line of charge with linear charge density λ , i.e. for each segment Δl of the line there is $\Delta Q = \lambda \Delta l$ of charge. Let's say we want to know the electric potential at some point in space with perpendicular distance to the line d - how do we compute that?

For the case of one point charge Q , we know that the electric potential is simply

$$\phi = \frac{kQ}{r}$$

but how do we extend this to an infinitely long line of charge?

We need the **principle of superposition**. The idea says that potentials can be added linearly, i.e. $\phi_{total} = \phi_1 + \phi_2 + \dots$

With this, we can break up the infinite line into infinitesimal segments of dx . If we do this, then each segment contributes

$$d\phi = \frac{k \cdot dq}{r}$$

Since the line is infinite, we can choose some point on the line as the origin of our coordinate system, which we'll choose to be the point right under the point of concern with perpendicular distance d .

$$d\phi = \frac{k \cdot \lambda dx}{\sqrt{x^2 + d^2}}$$

With this, we can write the total electric potential as:

$$\phi = \int d\phi = \int_{x=-\infty}^{x=+\infty} \frac{k\lambda dx}{\sqrt{x^2 + d^2}}$$

which one can compute with, say, a table of integrals.

4 Problems & Practices

4.1 Problem 1

Problem 1: An ideal gas is a volume of gas which obeys the equation of state

$$pV = nRT$$

where p is the pressure, V is the volume, n is the number of moles, T is the temperature in Kelvins and R is the molar gas constant. What is the total work done by the gas when it expands isothermally from V_i to V_f ?

Hint: Isothermal means that the temperature T is constant, and when an ideal gas expands by dV , it does work $W = pdV$ on its surroundings.

4.2 Problem 2

Problem 2: Derive the expression for the gravitational potential energy of a mass m , if the expression for the gravitational force is

$$\vec{F} = \frac{GMm}{r^2}(-\hat{r})$$

where M and m are masses and $\vec{r}$ is a vector pointing from the center of one mass to the other.

Note that the energy is defined to be zero at $r = \infty$.

Hint: Use the definition of work $dW = -dU = \vec{F} \cdot d\vec{r}$

4.3 Problem 3

Problem 3: Find the resistance of an Ohmic spherical shell with resistivity ρ , inner radius a and outer radius b .

Hint: the formula for resistance is $R = \frac{\rho L}{A}$, where L is the length of the resistor and A the cross-sectional area. Resistors in series obey the principle of superposition.

4.4 Problem 4

Problem 4: This problem deals with infinitesimals. Convince yourself that the following identity is true:

$$\ddot{x} = \dot{x} \frac{d\dot{x}}{dx}$$

4.5 Problem 5

Problem 5: In rotational mechanics, instead of using mass as a quantity for inertia, we use a quantity called the **moment of inertia**, often denoted I .

When a point mass m is spinning about some axis, its moment of inertia is

$$I = mr^2$$

where r is the perpendicular distance to the axis of rotation. And, just like mass in translational mechanics, the moment of inertia is additive.

With this information, derive the moment of inertia of a thin rod of length L and mass m spinning about its center.

4.6 Problem 6

Problem 6: The coordinates that we are familiar with are the xy coordinate system, also known as **rectangular coordinates**.

When we want to find the area of some patch of the grid, we can write an infinitesimal patch of area as

$$dA = dx \cdot dy$$

and integrate over the region of interest.

Another useful coordinate system is called **polar coordinates**. In this system, points in space are denoted with (r, θ) , where r is the distance from the origin, and θ the angle from some fixed axis.

Derive the formula for an infinitesimal patch of area dA in polar coordinates.

Hint: the area of a circular sector is $A = \frac{1}{2}r^2\theta$

4.7 Problem 7

Problem 7: When water in a container is rotated, its shape changes and curves at the edges. In this problem we derive the shape of the surface.

Consider a cylindrical container, rotating about its center. We let our origin be at the axis of rotation, at the cylinder's base. Let the density of water be ρ , gravitational acceleration be g , and the angular velocity of the rotation be ω .

- (a) Consider an infinitesimal blob of water at the base of the cylinder, and at some distance r from the axis of rotation. Write down the force balance equation for the blob to be in circular motion in the container.

Hint: think pressures!

- (b) Derive the pressure gradient, $\frac{dp}{dr}$ as a function of the distance r .
- (c) From this, derive the gradient of the surface of the water, $\frac{dy}{dr}$, if the surface of the water has the shape y as a function of r .
- (d) Derive the shape of the water surface from the result of (c).

Note: the result of d) implies that the surface of the liquid has a very well defined focal point - telescopes have been built by rotating liquids like this (**liquid-mirror telescopes**), like the Large Zenith Telescope!

5 Solutions

5.1 Problem 1

$$W = nRT \ln \frac{V_f}{V_i}$$

5.2 Problem 2

$$U = \int dU = - \int_{-\infty}^r \frac{GMm}{r^2} dr = - \frac{GMm}{r}$$

5.3 Problem 3

$$R = \int_a^b \frac{\rho dr}{4\pi r^2} = \frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$$

5.4 Problem 4

$$\frac{d^2x}{dt^2} = \frac{d\dot{x}}{dt} = \frac{d\dot{x}}{dx} \cdot \frac{dx}{dt} = \dot{x} \frac{d\dot{x}}{dx}$$

5.5 Problem 5

$$I = \int_{-L/2}^{L/2} r^2 dm = \int_{-L/2}^{L/2} x^2 \frac{M}{L} dx = \frac{1}{12} ML^2$$

5.6 Problem 6

$$dA = \frac{1}{2} r^2 d\theta$$

5.7 Problem 7

a)

$$p(r + dr)dA - p(r)dA = r\omega^2 dm$$

b)

$$p(r + dr)dA - p(r)dA = r\omega^2 \rho dA dr$$

$$\frac{dp}{dr} = \rho\omega^2 r$$

c)

$$\frac{dp}{dr} = \frac{\rho g dy}{dr}$$

$$\frac{dy}{dr} = \frac{\omega^2}{g} r$$

d) Integrating (c) gives

$$y = y_0 + \frac{\omega^2}{2g} r^2$$

6 Conclusion

Being acquainted with calculus-based physics is one of the big stepping stones on your way to becoming a theoretical physicist - congratulations on making it through this handout!

Taylor Series

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1 Introduction

Taylor Series are a remarkable piece of mathematical machinery that every aspiring theoretical physicist should know about! The reason they are so powerful is because their purpose is to transform an awkward function into a nice, polynomial one.

It is also useful in providing the rigor required to say that $\sin(x) \approx \tan(x) \approx x$, a common hand-wavy trick that makes physicists get on mathematicians' nerves.

Let's take a look at what the Taylor Series formally is!

2 Definition & Intuition

Definition: Given an n -time differentiable function $f(x)$, the polynomial

$$\begin{aligned} P_n(x) &= \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k \\ &= f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!} (x-a)^n \end{aligned}$$

is called a **Taylor Polynomial**, and its limit as you add infinitely many terms, $S(x)$

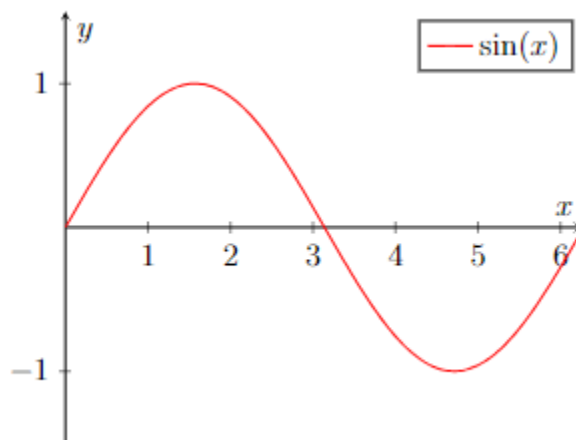
$$S(x) = \lim_{n \rightarrow \infty} P_n(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k$$

is called the **Taylor Series** of the function $f(x)$ centered around $x = a$.

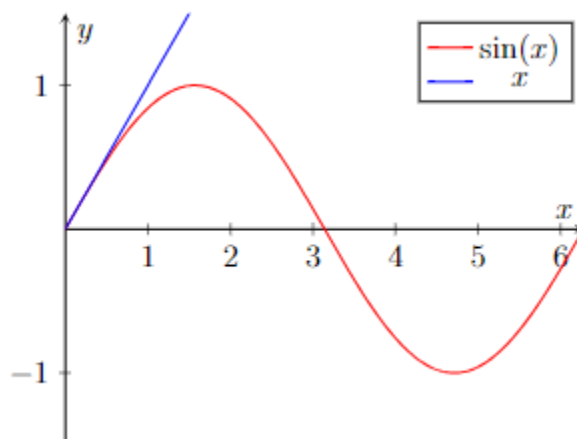
If $S(x)$ converges and is equal to the value of $f(x)$ everywhere, it is said that the **Taylor Expansion** of $f(x)$ is $S(x)$.

When $a = 0$ (i.e. the expansion is centered around the origin) the Taylor Expansion is called the **Maclaurin Expansion**.

To get an intuition of how this works, consider the function $f(x) = \sin(x)$. First, let's only look at values of x near the origin.



Close to the origin, the graph of $\sin(x)$ looks very similar to the linear function $y = x$, doesn't it? Let's plot both of them on the same graph.



Those two match really well near the origin! But, why is that?

The reason for their similarity is because their **first derivatives match** near the origin. When you differentiate $\sin(x)$, you get $\cos(x)$, and at $x = 0$ $\cos(x)$ is equal to 1. Hence the first derivative of $\sin(x)$ is 1 at the origin, which is same for the line $y = x$.

Now, can we improve on this? One thing we know about the sine function is that it is **odd**, that is, $\sin(-x) = -\sin(x)$. Hence, when we model them as a polynomial, the polynomial better be an odd function too - that is to say, $P_n(-x) = -P_n(x)$, which leads us to rule out all even powers of x (x^2 , x^4 , etc.).

Our next best guess at modelling the function is to add a downward-curving cubic term that matches the **higher derivatives of** $\sin(x)$. Let's call our new curve $y = x - cx^3$ for some constant c , as a guess. We put a negative because we need the line to curve downwards. Differentiating, we get

$$y' = 1 - 3cx^2$$

$$y'' = -6cx$$

$$y''' = -6c$$

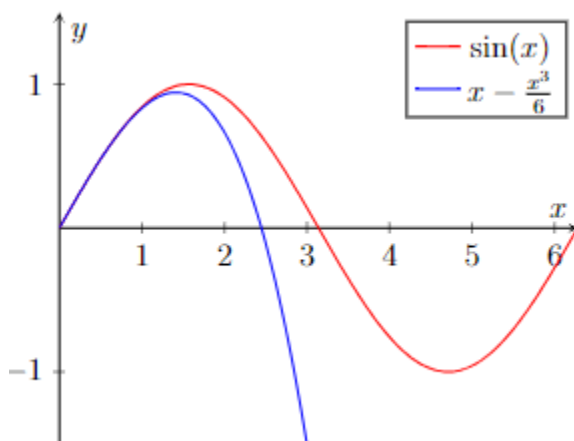
For the function $y = \sin(x)$, when we differentiate and plug in $x = 0$, we get

$$y' = \cos(0) = 1$$

$$y'' = -\sin(0) = 0$$

$$y''' = -\cos(0) = -1$$

For the third derivatives to match, we need $c = \frac{1}{6}$. Plotting this on the graph, we get:



And that's the core idea of Taylor series! When you do this same process over and over, matching higher and higher derivatives, the polynomial approaches the original function better and better. If instead of matching the function close to $x = 0$, you want to model the function around some point $x = a$, then you can just compute the derivatives at a and swap each power of x from x^k to $(x - a)^k$.

Feel free to play around with the idea in this Desmos plot.

Of course, there are some nuances about convergence and some oddly-behaving functions (see the problems below!), but let's leave that for the real-analysis mathematicians for now.

3 Common Taylor Series

3.1 Binomial Expansion

You might have learned this somewhere in school, and might be thinking "hey, this isn't a Taylor Series!" - you're totally right, but since these two ideas have many similar principles, it would be fitting to put them in the same discussion.

Definition: Given a binomial raised to a power, its expanded form can be written in the form

$$(x + a)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} x^k$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the **binomial coefficient**.

To see why this is true, let's see take a look when $x = 0$. On the RHS, all the terms would vanish except for the $k = 0$ term (because $0^0 = 1$, don't tell the mathematicians), leaving us with $\binom{n}{0}a^n = a^n$, which is exactly the same as the LHS.

Taking the derivatives of both sides, we get

$$n(x+a)^{n-1} = \sum_{k=0}^n \binom{n}{k} a^{n-k} k x^{k-1}$$

Again, at $x = 0$, the only nonzero term on the RHS is when $k = 1$, i.e. $\binom{n}{1}a^{n-1} = na^{n-1}$, which is also exactly the same as the LHS.

If you do this for all n higher derivatives, you would get that both sides always match, and hence the binomial expansion is true when viewed as a Taylor polynomial.

3.2 Trigonometric Functions

Here are a few classic trigonometric Taylor Expansions of trig functions. Try to convince yourself that they are true!

$$\begin{aligned}\sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \\ \cos(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k}\end{aligned}$$

You might be wondering what the Taylor expansion for $\tan(x)$ is, but that's a rabbit hole in and of itself, involving Bernoulli numbers from some magical math major's textbook. To expand $\tan(x)$ in Olympiad problems, just take the first few derivatives and forget about deriving a general expression for the expansion!

3.3 Exponential & Logarithmic Functions

Just as above, try convincing yourself that these equations hold true!

$$\begin{aligned}e^x &= \sum_{k=0}^{\infty} \frac{1}{k!} x^k \\ \ln\left(\frac{1}{1-x}\right) &= \sum_{k=1}^{\infty} \frac{1}{k} x^k\end{aligned}$$

Careful with the $\ln(x)$ function though! It starts at $k = 1$, because we don't want to be dividing by zero now do we...

You might notice that the form chosen for the Taylor expansion for $\ln(x)$ differs from other sources from the internet, but I find that this form is easier to remember.

3.4 Power Series

As with the Binomial Expansion, this one on the list isn't really a Taylor series either, but has significant overlap with the topic.

Definition: A power series $P(x)$ has the form

$$\begin{aligned} P(x) &= \sum_{k=0}^{\infty} a_k x^k \\ &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots \end{aligned}$$

where each of the coefficients a_k are constants.

A simple and familiar power series would be the power series of $\frac{1}{1-x}$, where all the constants a_k are just 1 because

$$\begin{aligned} \frac{1}{1-x} &= \sum_{k=0}^{\infty} x^k \\ &= 1 + x + x^2 + x^3 + \dots \end{aligned}$$

This is the generalized form of Taylor series, and can be found in many places of higher math & physics, ranging from generating functions used for combinatoric problems to solving the Schrodinger's equation for a Hydrogen-like atom.

One usually uses the power series to **construct a solution** to an equation, by plugging it into the equation and deducing what the coefficients must be.

An example is the familiar simple harmonic motion equation of motion:

$$y'' = -\omega^2 y$$

There are many ways to solve this, and one of them involves using power series.

Let $y = \sum_{k=0}^{\infty} a_k t^k$ be a solution to the SHM equation. Plugging it in, we get

$$\begin{aligned} \sum_{k=0}^{\infty} a_k \cdot k \cdot (k-1) \cdot x^{k-2} &= -\omega^2 \sum_{k=0}^{\infty} a_k t^k \\ \sum_{k=0}^{\infty} a_{k+2} \cdot (k+2) \cdot (k+1) \cdot x^k &= -\omega^2 \sum_{k=0}^{\infty} a_k t^k \end{aligned}$$

after some reindexing. Since the coefficients of x^k on each side must be identical, we obtain the relation

$$a_{k+2} = -\omega^2 \frac{1}{(k+2)(k+1)} a_k$$

Notice that this relation only talks about coefficients which are 2 indices away, so that means we have 2 families of a : one for odd k and one for even k , both determined by the initial conditions a_1 and a_0 respectively.

The recurrence relation reads

$$a_k = \frac{-\omega^2}{k(k-1)} a_{k-2}$$

$$a_k = \frac{-\omega^2}{k(k-1)} \cdot \frac{-\omega^2}{(k-2)(k-3)} a_{k-4}$$

$$a_k = \frac{-\omega^2}{k(k-1)} \cdot \frac{-\omega^2}{(k-2)(k-3)} \cdot \frac{-\omega^2}{(k-4)(k-5)} \cdots$$

We now split into two cases: k can either be even or odd.

When k is **even**, it can be written as $k = 2n$. We notice that when the recurrence relation is brought down to a_0 , there would be n copies of $-\omega^2$ in the numerator and $k!$ in the denominator, that is to say

$$a_{2n} = \frac{(-\omega^2)^n}{(2n)!} a_0$$

When k is **odd**, it can be written as $k = 2n + 1$. A similar argument would yield

$$a_{2n+1} = \frac{(-\omega^2)^n}{(2n+1)!} a_1$$

Finally, putting this into the original power series expression for y gives

$$y(t) = \sum_{\text{even}} a_k t^k + \sum_{\text{odd}} a_k t^k$$

$$y(t) = a_0 \sum_{n=0}^{\infty} \frac{(-\omega^2)^n}{(2n)!} t^{2n} + a_1 \sum_{n=0}^{\infty} \frac{(-\omega^2)^n}{(2n+1)!} t^{2n+1}$$

$$y(t) = a_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (\omega t)^{2n} + \frac{a_1}{\omega} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (\omega t)^{2n+1}$$

from which we recognize the two sums as

$$y(t) = a_0 \cos(\omega t) + \frac{a_1}{\omega} \sin(\omega t)$$

4 Applications

4.1 Approximations

Let's go back to the power series of $\frac{1}{1-x}$, where all the terms in the expansion differs by a factor of x . If x is a very small number, eg. 0.01, we would find that

n	x^n	$P_n(0.001)$	$P_n(x)$
0	1	1	1
1	0.1	1.001	$1 + x$
2	0.01	1.001001	$1 + x + x^2$
3	0.001	1.001001001	$1 + x + x^2 + x^3$

The values that add onto the Taylor polynomial become smaller and smaller. In physics, one philosophy that one must adhere to is that **nothing is perfect**. As said by Greg in the hit show Steven Universe: “if all porkchops were perfect, we wouldn’t have hotdogs!” If all meter rulers were perfect, we wouldn’t have vernier calipers, micrometers, etc. The thing is, only up to a certain precision is a number useful for. Beyond that, we can never measure something that precisely, and the digits would mean nothing.

All of this is to say that the expression $\frac{1}{1-x}$ can be approximated as $1+x$, as that’s the useful range of precision we could hope for. (in different scenarios, the range of precision would be different too, but that’s up to the physicist to be the judge of what precision is necessary).

Some other famous approximations would include

$$\sin(x) \approx x$$

$$\cos(x) \approx 1 - \frac{x^2}{2}$$

$$e^x \approx 1 + x$$

$$(1+x)^n \approx 1 + nx$$

all of which only hold true when $|x| \ll 1$

4.2 SHM angular frequency

We know that the simple harmonic motion equation of motion has the form

$$\ddot{x} = -\omega^2 x$$

where ω is the angular frequency of the system, however the system is setup. What if we want to generalize this and find ω knowing only the potential energy U ?

Suppose we have a potential energy function $U(x)$, and the system oscillates about an equilibrium point x_0 . You can imagine a mass on a spring oscillating under the influence of gravity - the equilibrium point is somewhere lower than the spring’s natural length. At this point

Most forces we deal with have a special property, that is that they are **conservative** and obey the equation $F = -\frac{dU}{dx}$. One can see this from the fact that work, the change in energy, is defined as $W = \vec{F} \cdot \vec{\Delta s}$.

We know that, by Newton’s 2nd law, $F = m\ddot{x}$, hence

$$-\frac{dU}{dx} = m\ddot{x}$$

Here comes the magic. If we do a Taylor expansion of the potential energy function about the point x_0 , one gets

$$U(x) \approx U(x_0) + U'(x_0)\frac{(x-x_0)}{1!} + U''(x_0)\frac{(x-x_0)^2}{2!}$$

from neglecting the higher order terms by assuming that the oscillations are small.

From here, we get the expression for $\frac{dU}{dx}$ as

$$\frac{dU}{dx} \approx U'(x_0) + U''(x_0)(x - x_0)$$

$$\frac{dU}{dx} \approx U''(x_0)(x - x_0)$$

because by definition of being an equilibrium point, $U'(x_0) = F_{x=x_0} = 0$.

If we let the perturbation from equilibrium $x - x_0$ be ϵ , then $\ddot{x} = \ddot{\epsilon}$. Putting all these into the original equation of motion, we get

$$\begin{aligned} -\frac{dU}{dx} &= m\ddot{x} \\ -U''(x_0)\epsilon &= m\ddot{\epsilon} \\ \ddot{\epsilon} &= -\frac{U''(x_0)}{m}\epsilon \end{aligned}$$

which is exactly the form of SHM. Directly comparing coefficients, we get

$$\omega = \sqrt{\frac{U''(x_0)}{m}}$$

5 Problems & Practices

5.1 Problem 1

Problem 1: Show that both of the following "famous" definitions of e^x are equivalent:

$$\begin{aligned} e^x &= \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n \\ e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \end{aligned}$$

(Note: these are the real definitions of e^x - 2.71828... is not just a number!)

5.2 Problem 2

Problem 2: Find the Taylor expansion of $\arctan x$.

5.3 Problem 3

Problem 3: From the Desmos plot earlier in this handout, you played around with the function $\sin(x)$, now let's take a look at the expansion of $\ln\left(\frac{1}{1-x}\right)$

- (a) Change the function in the Desmos to the logarithmic function. Does the Taylor expansion match the function well? What if you add more and more terms?

- (b) Take a look at the coefficients in the Taylor expansion

$$\ln\left(\frac{1}{1-x}\right) = \sum_{k=1}^{\infty} \frac{1}{k} x^k$$

Under what circumstances will the series converge? (hint: think ratios)

- (c) Obtain the range of values x on which the infinite series converges. This result is called the **radius of convergence**, and is the range where the Taylor expansion holds. (careful about the inequality boundaries!)

5.4 Problem 4

Problem 4: Prove Euler's Formula,

$$e^{ix} = \cos(x) + i \sin(x)$$

where i is the imaginary number which obeys $i^2 = -1$.

5.5 Problem 5

Problem 5: Suppose there are two electric charges q and $-q$, respectively, at $(0, d)$ and $(0, -d)$ in vacuum. Derive the potential V at (x, y) , which is located sufficiently far from the origin, up to the first order with respect to d . Let the potential at infinity be zero, and denote the vacuum permittivity by ϵ_0 .

Hint: The potential of one electric charge Q at a distance r is $V = \frac{Q}{4\pi\epsilon_0 r}$, and potentials obey the principle of superposition (i.e. $V_{\text{tot}} = V_1 + V_2$)

5.6 Problem 6

Problem 6: In the presence of a uniform and weak magnetic field B pointing into the page, a particle of charge q and mass m with velocity v is shot to the right toward a point P on the screen which is located at distance L away from the initial position of the particle. The incident particle is moving perpendicularly to the screen.

Derive the distance l between P and the point where the particle actually hits the screen, assuming that the deflection angle of the particle (which indicates the change in the advancing direction) is sufficiently small.

Hint: the particle undergoes circular motion under the centripetal force supplied by the Lorentz force, $F = q\vec{v} \times \vec{B}$.

5.7 Problem 7

Problem 7: You might have heard of the James Webb Space Telescope (JWST) that's been supplying the pretty images we see on social media, but did you know that its position in space is a very special one?

It orbits in space at a special point: the **Lagrange point** (specifically L2), which moves in tandem with the Earth, orbiting along the same line joining the Earth and the Sun. (for a better visualisation, search up some GIFs on Google!)

Derive the distance between the L2 point, at which JWST is currently orbiting around, and the Earth.

Hint: The reason it can achieve this special configuration is because the gravitational force from both the Sun and the Earth perfectly supply the centripetal force needed for JWST to orbit at that particular point.

Bonus: if you're up for a challenge, try your hand at IPhO 2011 Q1!

6 Solutions

6.1 Problem 1

Applying binomial theorem on the first definition and simplifying would give the second definition.

6.2 Problem 2

Differentiate repeatedly to get

$$\arctan x = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1}$$

6.3 Problem 3

a) Diverges at the sides, only converges on some interval

b) When the terms you add get smaller and smaller, i.e.

$$\left| \frac{T_{n+1}}{T_n} \right| < 1$$

$$\left| \frac{\frac{1}{n+1}x^{n+1}}{\frac{1}{n}x^n} \right| < 1$$

$$\left| \frac{n}{n+1}x \right| < 1$$

c) The inequality derived above must hold even when n is infinitely large

$$\lim_{n \rightarrow \infty} \left| \frac{n}{n+1}x \right| < 1$$

$$|x| < 1$$

$$-1 < x < 1$$

But notice that we can also include $x = -1$ too because

$$\ln 2 = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

Hence

$$-1 \leq x < 1$$

6.4 Problem 4

Use Taylor expansion of e^x and change variables $x \rightarrow ix$, then separate real and imaginary parts to get Taylor series $\cos x$ and $\sin x$.

6.5 Problem 5

For some point in space (x, y) the potential is

$$V = \frac{kq}{\sqrt{x^2 + (y-d)^2}} - \frac{kq}{\sqrt{x^2 + (y+d)^2}}$$

To simplify this, we can write

$$\frac{1}{\sqrt{x^2 + (y \pm d)^2}} = \frac{1}{\sqrt{x^2 + y^2}} \left(1 + \frac{\pm 2yd + d^2}{x^2 + y^2} \right)^{-\frac{1}{2}}$$

And using $(1+x)^n \approx 1+nx$ for small x we get

$$\begin{aligned} \frac{1}{\sqrt{x^2 + (y \pm d)^2}} &\approx \frac{1}{\sqrt{x^2 + y^2}} \left(1 - \frac{\pm 2yd + d^2/2}{x^2 + y^2} \right) \\ &\approx \frac{1}{\sqrt{x^2 + y^2}} \left(1 \mp \frac{yd}{x^2 + y^2} \right) \end{aligned}$$

where we discarded the second order term of d . Therefore the potential is

$$V = \frac{kq \cdot 2yd}{(x^2 + y^2)^{\frac{3}{2}}}$$

6.6 Problem 6

Under the influence of a magnetic field a charged particle undergoes circular motion with radius

$$r = \frac{mv}{qB}$$

From geometry,

$$l = r - r \cos \theta \approx r \frac{\theta^2}{2}$$

Also from geometry,

$$\sin \theta = \frac{L}{r} \approx \theta$$

Hence

$$l \approx r \frac{\left(\frac{L}{r}\right)^2}{2} = \frac{L^2}{2r} = \frac{qBL^2}{2mv}$$

6.7 Problem 7

At the Lagrange point L2 force balance gives

$$\frac{GM_S}{(R+d)^2} + \frac{GM_E}{d^2} = (R+d)\omega^2$$

where ω is the angular velocity of the Earth's revolution around the Sun. This can also be expressed with Kepler's 3rd Law:

$$GM_S = \omega^2 R^3$$

Substituting that into the original force balance equation,

$$\begin{aligned} \frac{GM_S}{R^2} \left(1 + \frac{d}{R}\right)^{-2} + \frac{GM_E}{d^2} &= R \left(1 + \frac{d}{R}\right) \frac{GM_S}{R^3} \\ \frac{GM_S}{R^2} \left(1 - 2\frac{d}{R}\right) + \frac{GM_E}{d^2} &= \frac{GM_S}{R^2} \left(1 + \frac{d}{R}\right) \\ -2\frac{GM_S d}{R^3} + \frac{GM_E}{d^2} &= \frac{GM_S d}{R^3} \\ \frac{M_E}{d^3} &= \frac{3M_S}{R^3} \\ d &= R \left(\frac{M_E}{3M_S}\right)^{\frac{1}{3}} \end{aligned}$$

IPhO 2011 Q1 solution can be found [here](#).

7 Conclusion

Understanding Taylor Series and its power in simplifying equations is your first step in becoming a mathematically well-equipped theoretical physicist! Congratulations on making it through this handout!

L'Hôpital's Rule

Abheek Dhawan

July 2023

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1 Introduction

L'Hôpital's Rule is an important technique that can be used to make our lives easier when dealing with certain forms of tricky limits – those that evaluate to indeterminate form.

2 Identifying Indeterminate Form

2.1 Indeterminate Form

Indeterminate form applies when evaluating a limit leads to an expression that cannot be truly evaluated. While indeterminate form is stated singularly, there are actually many different kinds of limits that satisfy this categorization.

2.1.1 Quotient Forms

Quotient forms are the most common type of indeterminate form. These include the form $\frac{0}{0}$ and $\frac{\infty}{\infty}$.

2.1.2 Other Forms

- **Product Form:** characterized by $0 \cdot \infty$ (e.g. $\lim_{x \rightarrow 0^+} x \ln x$)
- **Difference Form:** characterized by $\infty - \infty$ (e.g. $\lim_{x \rightarrow 0^+} \frac{1}{x^2} - \frac{1}{\tan x}$)
- **Power Form:** characterized by ∞^0 , 0^0 , and 1^∞ (e.g. $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$)

2.2 Getting to Quotient Form

To be useful for L'Hôpital's Rule, it is necessary to get a limit from whatever form it is in to begin with into some sort of quotient. As there is no hard-and-fast rule to do this, some examples will be presented in Section 3.

3 Applying L'Hôpital's Rule

Applying L'Hôpital's Rule is relatively straightforward once the limit has been identified and manipulated to be in quotient form. The rule is defined with a limit in the following form:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

If the fraction $\frac{f(x)}{g(x)}$ evaluates to either $\frac{0}{0}$ or $\frac{\infty}{\infty}$, the limit can be properly evaluated by replacing the quotient with

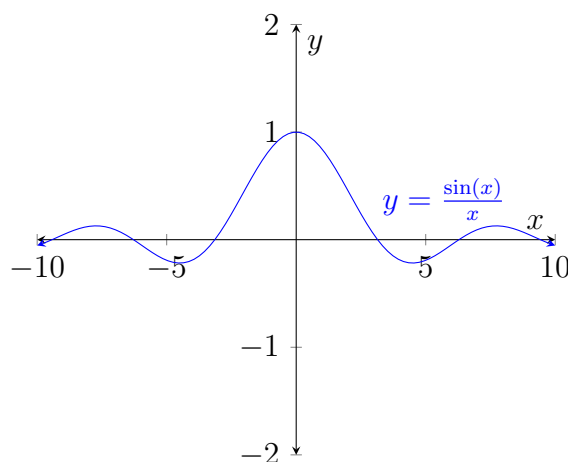
Definition of L'Hôpital's Rule

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

If this results in indeterminate form yet again, L'Hôpital's Rule can be used repetitively.

3.1 Example: $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$

A common and straightforward example to demonstrate a use of L'Hôpital's Rule can be found in $\frac{\sin(x)}{x}$.



As can be seen from the graph, $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$ is clearly 1. However, when evaluating the limit algebraically, it can be seen that both $\sin(x)$ and x evaluate to 0, leaving the indeterminate form $\frac{0}{0}$. Armed with the knowledge of L'Hôpital's Rule, we can set $f(x)$ to $\sin(x)$ and $g(x)$ to x . Evaluating $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$ leaves us with a limit $\lim_{x \rightarrow 0} \frac{\cos(x)}{1} = 1$, agreeing with the graphical approach. This example represents an extremely common and invaluable use case of L'Hôpital's Rule.

3.2 Example: $\lim_{x \rightarrow \infty} \frac{3x+5}{2x+1}$

Another relatively common example is having to evaluate a limit as x approaches ∞ . In this case, we have the limit

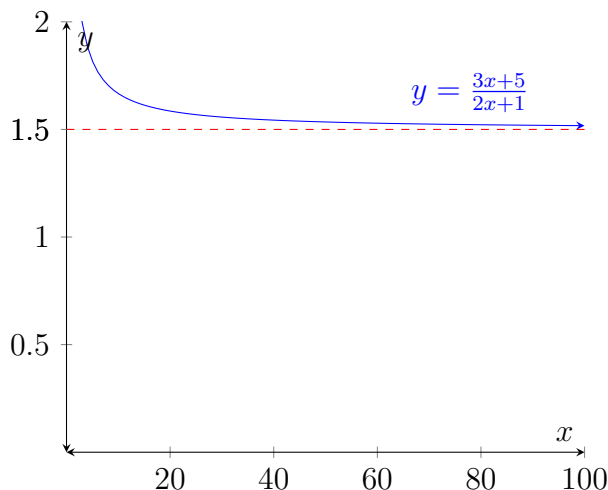
$$\lim_{x \rightarrow \infty} \frac{3x+5}{2x+1}$$

Applying L'Hôpital's Rule:

$$\lim_{x \rightarrow \infty} \frac{3}{2}$$

There is nothing left to plug in, and the limit therefore evaluates to $\frac{3}{2} = 1.5$.

Let's take a look at the graph to see this visually:



3.3 Example: $\lim_{x \rightarrow 0^+} x \ln x$

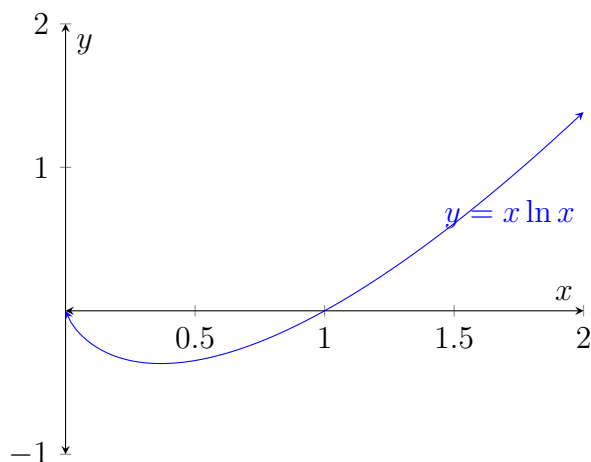
An example for the form $0 \cdot \infty$ can be found in $\lim_{x \rightarrow 0^+} x \ln x$. We can rearrange this into a quotient by dividing by $\frac{1}{x}$ instead of multiplying by x :

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x}$$

The derivative of the numerator is $\frac{1}{x}$ and that of the denominator is $-\frac{1}{x^2}$. Applying L'Hôpital's Rule, the limit becomes

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} \\ \lim_{x \rightarrow 0^+} -x \\ = 0 \end{aligned}$$

Therefore, it can be seen the limit evaluates to 0. Let's confirm graphically:



3.4 Example: $\lim_{x \rightarrow 0^+} \frac{1}{x^2} - \frac{1}{\tan x}$

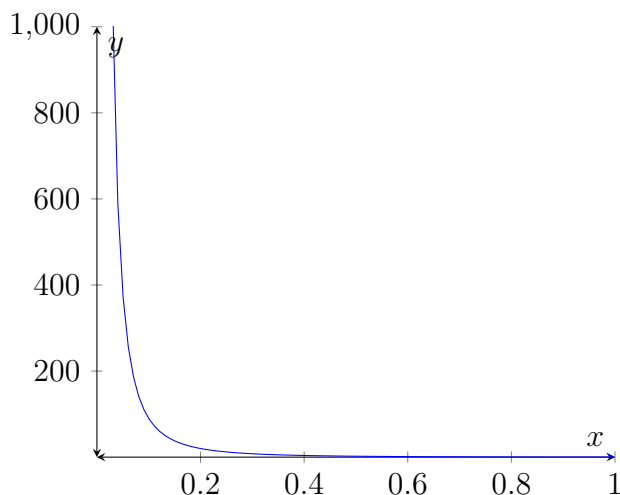
In this case we will be evaluating a limit of the indeterminate form $\infty - \infty$: $\lim_{x \rightarrow 0^+} \frac{1}{x^2} - \frac{1}{\tan x}$. As x goes to 0 both denominators do as well, and since the numerator is nonzero both terms approach ∞ , requiring another method – L'Hôpital's Rule. The limit must first be converted to a quotient, which can be done by multiplying each term by the other term's denominator:

$$\lim_{x \rightarrow 0^+} \frac{\tan(x) - x^2}{x^2 \tan(x)}$$

Taking the necessary derivatives:

$$\lim_{x \rightarrow 0^+} \frac{\sec^2(x) - 2x}{x^2 \sec^2(x) + 2x \tan(x)}$$

As x approaches 0, $\sec^2 x$ approaches 1, and $2x$ approaches 0, leaving the numerator to approach 1 as a whole. The denominator, however, approaches 0, leaving the final limit to approach ∞ . Again, let's confirm this graphically:



3.5 Example: $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$

This kind of limit is a bit more complex to evaluate. When we initially attempt to evaluate it algebraically, we get ∞^0 , which has no definite solution. To apply L'Hôpital's Rule, we must first create a quotient. This can be done by taking the natural log:

$$\ln x^{\frac{1}{x}} = \frac{\ln x}{x}$$

However, we have now modified the original limit. We can use a simple characteristic of continuous functions

$$\lim_{x \rightarrow a} \ln f(x) = \ln(\lim_{x \rightarrow a} f(x))$$

In this case $f(x)$ is the original expression, $x^{\frac{1}{x}}$. The previous equation then evaluates to

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \ln\left(\lim_{x \rightarrow \infty} x^{\frac{1}{x}}\right)$$

Applying L'Hôpital's Rule:

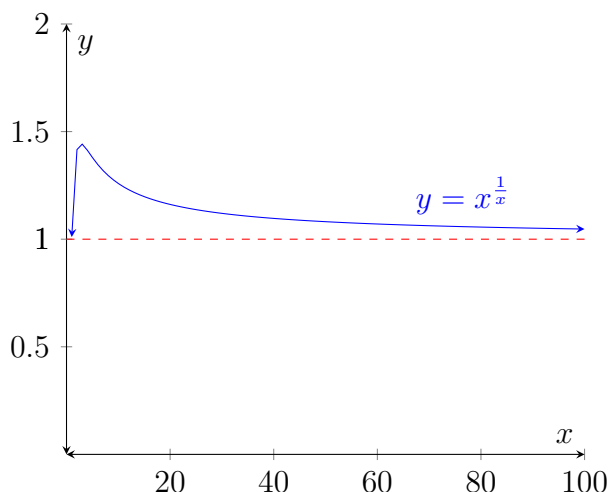
$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

Using the previous equation:

$$\ln\left(\lim_{x \rightarrow \infty} x^{\frac{1}{x}}\right) = 0$$

$$\lim_{x \rightarrow \infty} x^{\frac{1}{x}} = 1$$

Let's confirm this graphically:



3.6 Proof

We can begin with a simple quotient limit:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

If the limit evaluated algebraically leads to the form $\frac{0}{0}$ then $f(a) = g(a) = 0$.

$$\begin{aligned} &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} \\ &= \lim_{x \rightarrow a} \frac{\frac{f(x) - f(a)}{x - a}}{\frac{g(x) - g(a)}{x - a}} \\ &= \frac{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}}{\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}} \end{aligned}$$

The numerator and denominator can now be seen to be the difference quotients $f'(x)$ and $g'(x)$. This leaves the final result when $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ evaluates to indeterminate form to be:

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

3.7 Shortcomings

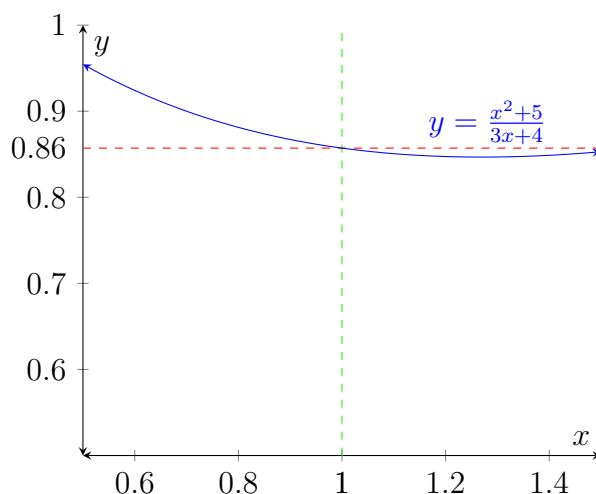
Let's take an example where the result is not indeterminate form to prove that L'Hôpital's Rule works exclusively when indeterminate form is encountered.

$$\lim_{x \rightarrow 1} \frac{x^2 + 5}{3x + 4}$$

When evaluating algebraically, the result comes out to $\frac{6}{7}$. Let's see what happens if we try to apply L'Hôpital's Rule.

$$\lim_{x \rightarrow 1} \frac{2x}{3}$$

When attempting to evaluate this, we get a different result of $\frac{2}{3}$, which is clearly incorrect if it is continuous at $x = 1$. This becomes even more apparent when looking at the graph:



4 Conclusion

L'Hôpital's Rule is an extremely useful tool which allows to solve many more limits than would've been possible without it. It's simplicity further contributes to it being a useful part of our calculus arsenal.

5 Afterword

Now that you know all there is to know about L'Hôpital's Rule, please enjoy this comic I made sophomore year:

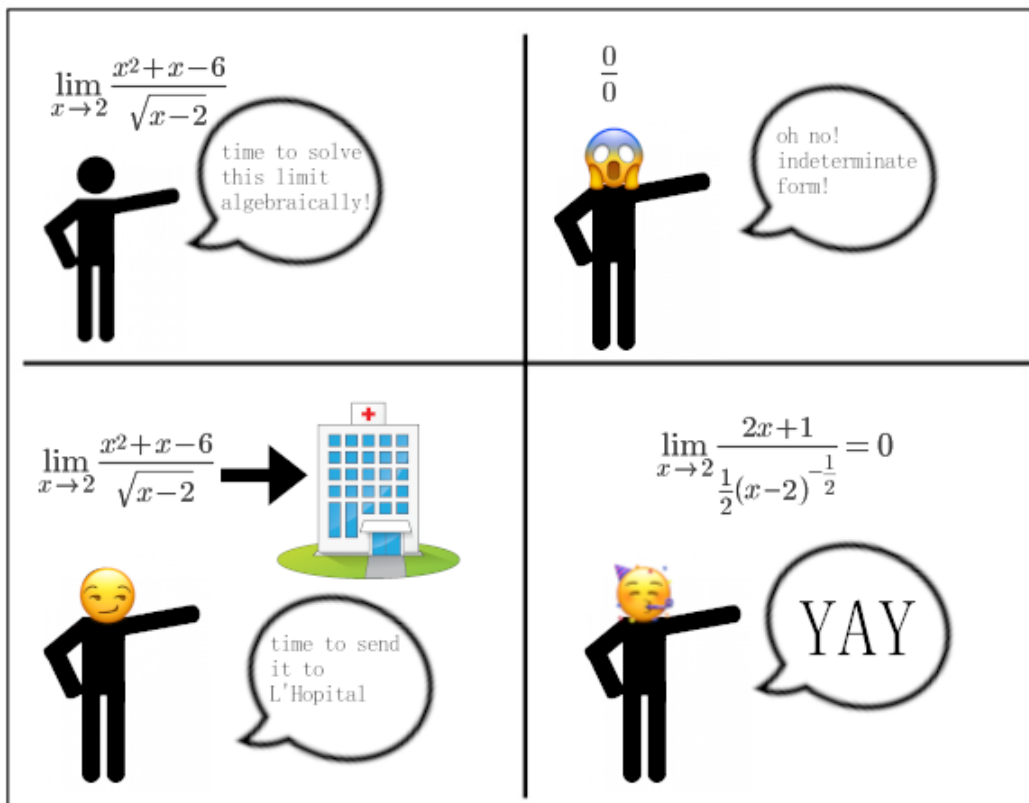


Figure 1: Cool guy saved by awesome application of L'Hôpital's Rule.

Optimization

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August 2023

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1 Introduction

Optimization is a fundamental concept in calculus that plays a crucial role in physics. It involves finding the best or optimal solution to a problem, given certain constraints or conditions.

2 Understanding Optimization Problems

2.1 Definition of Optimization

Optimization is the process of maximizing or minimizing a function's value, subject to a set of constraints. In mathematical terms, an optimization problem involves finding the values of variables that optimize the objective function while satisfying the given constraints. The objective function represents the quantity to be optimized, and the constraints define the conditions that the solution must adhere to.

2.2 Objective Function and Constraints

The objective function is a mathematical representation of the quantity that you want to optimize. It can be expressed as a function of one or more variables. The goal is to find the values of these variables that maximize or minimize the objective function.

Constraints, on the other hand, are conditions or limitations that the solution must satisfy. These constraints can be in the form of equations or inequalities and may impose restrictions on the values of the variables.

2.3 Examples

To better understand optimization problems, let's consider some examples:

Maximizing the area of a rectangle with a fixed perimeter.

In this example, suppose you have a fixed length of fencing material and you want to construct a rectangular enclosure. The goal is to find the dimensions of the rectangle that maximize its area. By maximizing the area, you can make the most efficient use of the available fencing material.

Minimizing the cost of producing a certain quantity of goods.

In this example, consider a manufacturing company that produces a certain quantity of goods. The cost of production depends on various factors, such as raw material costs, labor costs, and overhead expenses. The objective is to determine the optimal production quantity that minimizes the overall cost of production. This optimization problem involves minimizing the cost function (the objective function) while satisfying production constraints, such as resource limitations or demand requirements.

Maximizing the profit from selling a product with limited resources.

3 Techniques for Solving Optimization Problems

3.1 Critical Points

In optimization, critical points are the values of the variables where the objective function's derivative is either zero or undefined. These points are important because they can correspond to maximum or minimum values of the objective function.

3.2 The First Derivative Test

The first derivative test is a technique used to determine whether a critical point corresponds to a **relative maximum**, **relative minimum**, or neither. The first derivative indicates if the function is increasing, decreasing, or neither. If the first derivative is positive, the function is increasing, and if it is negative, the function is decreasing.

We take the first derivative of the objective function and set that function equal to zero or undefined, then solve for the x values which will help us find the critical point. Once we find the x values which make the derivative 0, we check the interval to the left and to the right of the critical point. We can check the interval to the left and right of the critical points by plugging values into the derivative. If the derivative is going from positive to negative, the critical point is a max, and if the derivative is going from negative to positive, the critical point is a min. If we are looking for the **absolute maximum**, we find the maximum of all the relative maximum points we calculated, and vice versa.

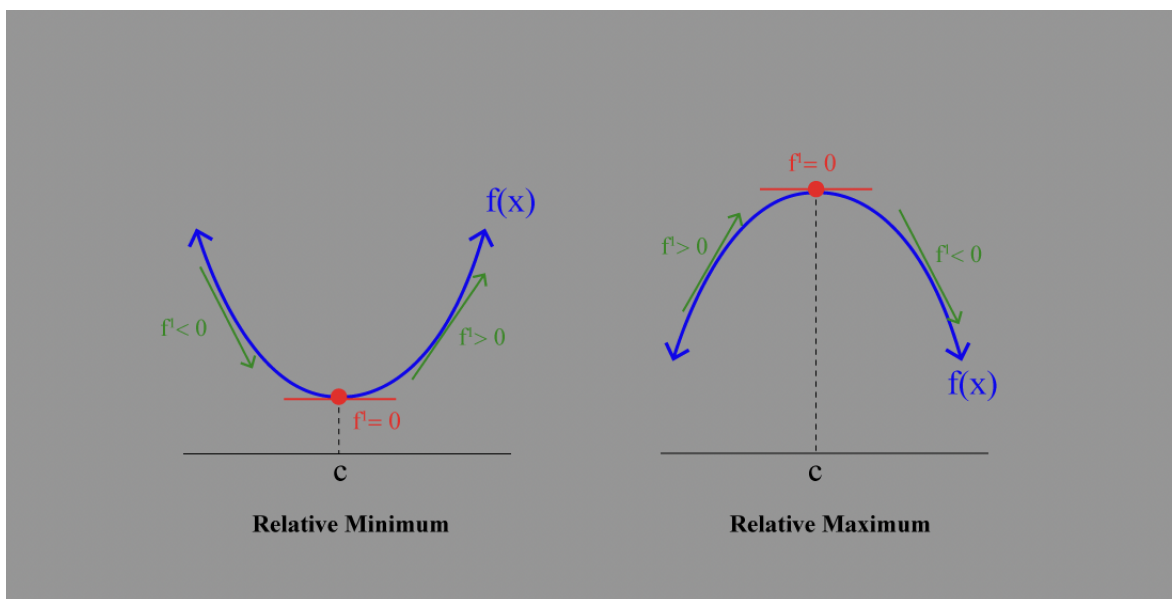


Figure 1: Relative min and max (source: Ximera)

3.3 The Second Derivative Test

The second derivative test is another method for identifying the nature of critical points. The second derivative indicates the concavity of a function. If the second derivative is positive over a certain interval, then the function is concave up of that interval, and if it is negative, then the function is concave down. If the second derivative is zero, we will not have to worry about the x value because it will be an inflection point, and inflection points can never be a maximum or minimum, since concavity changes at inflection points.

We check for critical points by setting the first derivative function equal to zero. Instead of checking the intervals to the left and right of the critical point, we can check if the second derivative is positive or negative at that point, to determine if that point is a relative max or min. We then do the same process, where we find the maximum over all relative maximums or vice versa to find the absolute max or min.

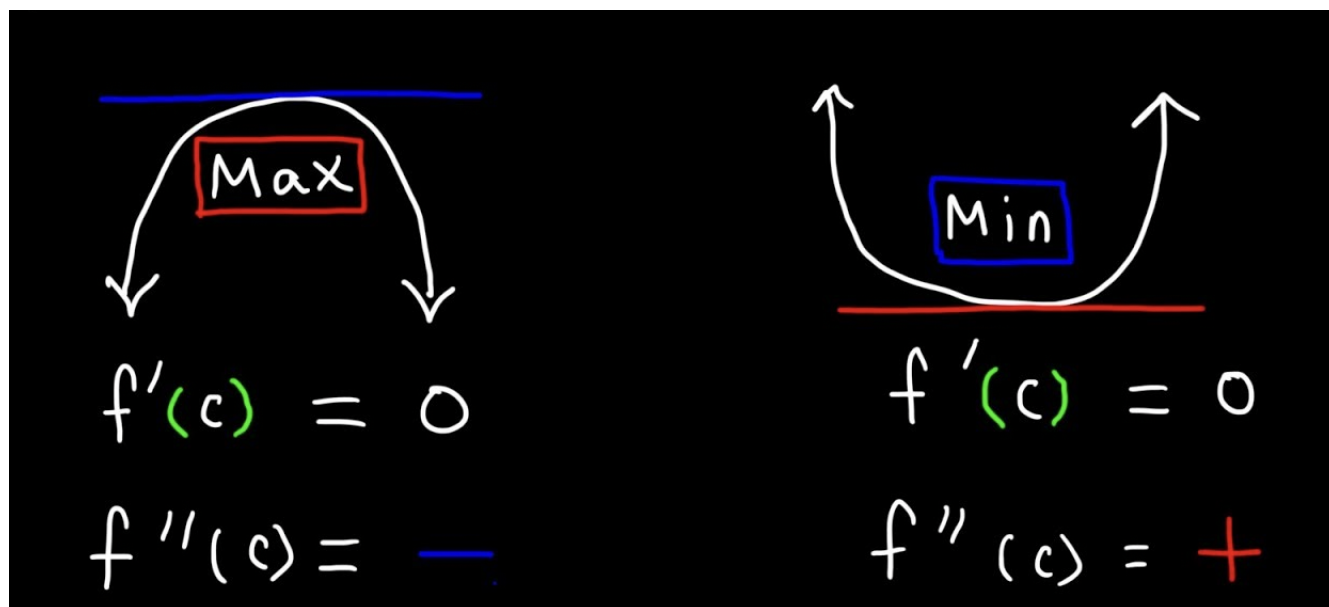


Figure 2: Second Derivative Test (source: Organic Chemistry Tutor)

4 Practice Problems

To reinforce your understanding of optimization, here are some practice problems that you can solve:

4.1 Problem 1

A rectangular garden is to be constructed using a rock wall as one side of the garden and wire fencing for the other three sides. Given 100ft of wire fencing, determine the dimensions that would create a garden of maximum area. What is the maximum area? (Source: Openstax)

Solution: Let's assume the length of the rock wall is L and the width of the garden, which will be enclosed by wire fencing, is W . The total length of wire fencing used is given by:

$$2W + L = 100$$

To maximize the area of the garden, we need to find the dimensions that maximize $A = LW$. By solving the equation $2W + L = 100$ for L , we get:

$$L = 100 - 2W$$

Substituting this into the area equation, we have:

$$A = (100 - 2W)W = 100W - 2W^2$$

To find the maximum area, we take the derivative of A with respect to W and set it equal to zero:

$$\frac{dA}{dW} = 100 - 4W = 0$$

Solving this equation gives us $W = 25$.

Taking the second derivative of A with respect to W , we have:

$$\frac{d^2A}{dW^2} = -4$$

Since the second derivative is negative, this indicates a concave-down function, which means that there exists a maximum at $W = 25$.

Substituting $W = 25$ back into $2W + L = 100$, we find $L = 50$.

Therefore, the dimensions that create a garden of maximum area are:

$$W = 25 \text{ ft} \quad \text{and} \quad L = 50 \text{ ft}$$

Substituting these values back into the area equation, we calculate the maximum area:

$$A = LW = 25 \text{ ft} \times 50 \text{ ft} = 1250 \text{ square feet}$$

4.2 Problem 2

Which point on the graph of $y = x$ is closest to the point $(5, 0)$?

Solution: To find the point on the graph of $y = x$ that is closest to the point $(5, 0)$, we can use calculus and the concept of minimizing the distance between two points.

Let's denote the point on the graph of $y = x$ that is closest to $(5, 0)$ as (x, x) . The distance between these two points can be expressed as:

$$d = \sqrt{(x - 5)^2 + (x - 0)^2}$$

To find the minimum distance, we need to minimize the function d . We can do this by finding the critical points of d by taking its derivative and setting it equal to zero:

$$\frac{d}{dx} = \frac{2(x-5) + 2(x-0)}{2\sqrt{(x-5)^2 + (x-0)^2}} = 0$$

Simplifying the equation, we have:

$$2(x-5) + 2x = 0$$

Solving for x , we find:

$$4x - 10 = 0 \implies x = \frac{10}{4} = \frac{5}{2}$$

Thus, the critical point occurs at $x = \frac{5}{2}$. To determine whether this point corresponds to a minimum or maximum, we can perform the first derivative test. We evaluate the first derivative of d at a value slightly less than $\frac{5}{2}$ and a value slightly greater than $\frac{5}{2}$:

$$\left. \frac{d}{dx} \right|_{x=\frac{5}{2}^-} = \frac{2(\frac{5}{2}-5) + 2(\frac{5}{2}-0)}{2\sqrt{(\frac{5}{2}-5)^2 + (\frac{5}{2}-0)^2}} < 0$$

$$\left. \frac{d}{dx} \right|_{x=\frac{5}{2}^+} = \frac{2(\frac{5}{2}-5) + 2(\frac{5}{2}-0)}{2\sqrt{(\frac{5}{2}-5)^2 + (\frac{5}{2}-0)^2}} > 0$$

Since the derivative changes sign from negative to positive at $x = \frac{5}{2}$, this indicates that $x = \frac{5}{2}$ corresponds to a minimum point. Since $\frac{5}{2}$ is the only relative minimum, it is the absolute maximum.

Therefore, the point on the graph of $y = x$ closest to $(5, 0)$ is $(\frac{5}{2}, \frac{5}{2})$.

4.3 Problem 3.

An open box is to be made from a 16in by 30in piece of cardboard by cutting out squares of equal size from the four corners and bending up the sides. What size should the squares be to obtain a box with the largest volume?

Solution: Let's assume that the side length of the squares to be cut out from each corner is x (in inches).

After cutting out the squares and bending up the sides, the length of the resulting box will be $30 - 2x$ inches, the width will be $16 - 2x$ inches, and the height (the side length of the squares) will be x inches.

The volume V of the box can be calculated by multiplying the length, width, and height:

$$V = (30 - 2x)(16 - 2x)x$$

Simplifying the expression, we have:

$$V = 4x^3 - 92x^2 + 480x$$

To find the size of the squares that maximizes the volume, we need to find the critical points of V . We can do this by taking the derivative of V with respect to x and setting it equal to zero:

$$\frac{dV}{dx} = 12x^2 - 184x + 480 = 0$$

Solving this quadratic equation for x , we can use factoring, completing the square, or the quadratic formula.

In this case, factoring the quadratic equation gives us:

$$4(3x - 10)(x - 12) = 0$$

This gives us two critical points: $x = \frac{10}{3}$ and $x = 12$. We reject $x = 12$ because $0 \leq x \leq 8$. This happens because if x exceeds 8, then we are unable to cut squares out of the corners, because two squares that share the shorter edge will exceed the value of 16.

To determine which value of x gives the maximum volume, we can evaluate the second derivative of V :

$$\frac{d^2V}{dx^2} = 24x - 184$$

Substituting $x = \frac{10}{3}$ into the second derivative, we find:

$$\left. \frac{d^2V}{dx^2} \right|_{x=\frac{10}{3}} = 24 \left(\frac{10}{3} \right) - 184 = -8 < 0$$

Since the second derivative is negative for $x = \frac{10}{3}$ this confirms that $x = \frac{10}{3}$ gives the maximum volume as it is the only relative max we have.

Therefore, to obtain a box with the largest volume, the size of the squares to be cut out from each corner should be $\frac{10}{3}$ inches, resulting in a maximum volume of 725.926 in^3 .

Yashpatil < @bohring_bot

Related Rates

Ashwin Narayanan

July 2023

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1 Introduction

Related rates is a fascinating branch of calculus that deals with the study of how different quantities change in relation to each other. It is used to solve real-world problems where multiple variables are interdependent and changing with respect to time. By examining the rate of change of one quantity, we can determine how other related quantities change simultaneously.

In related rates problems, we analyze the relationships between variables using derivatives. The process typically involves identifying relevant equations, differentiating them with respect to time (using the chain rule), and then solving for the desired rate of change. Practice is key to mastering related rates problem-solving, and I highly encouraged to draw diagrams and label variables to effectively understand the problem.

1.1 Key Steps

Read the following steps carefully and identify the changing variables and the given rates of change.

1. **Identify the Variables:** Begin by identifying all the variables involved in the problem. Assign meaningful names to each quantity and denote their respective rates of change with respect to time.
2. **Establish an Equation:** Next, establish an equation that relates the variables. This equation usually stems from the problem's context or geometry.
3. **Differentiate with Respect to Time:** Differentiate the equation with respect to time (t) to obtain a new equation that relates the rates of change of the variables.
4. **Plug in Known Values:** Substitute the known values of the variables and their rates into the derived equation.
5. **Solve for the Desired Rate:** Solve for the unknown rate of change, which is usually the rate you are asked to find in the problem.

Example 1: Consider a streetlight mounted on top of a pole and a person walking away from the pole. The person's height is 1.8 meters, and their walking speed is 1.5 m/s. When the person is 10 meters away from the pole, the person's shadow is 6 meters long. At what rate is the tip of the person's shadow moving concerning time when they are 10 meters away from the pole?

Solution: Given:

- Let h be the height of the person (constant) = 1.8 meters.
- Let x be the distance between the person and the pole, changing with time (in meters).
- Let s be the length of the person's shadow, changing with time (in meters).

From the given information, we have $h = 1.8$ m and $\frac{dx}{dt} = -1.5$ m/s (negative since the person is walking away). Using similar triangles, we can relate the variables x and s :

$$\frac{s}{x} = \frac{h}{10}$$

Now, differentiate both sides of the equation with respect to time t using the chain rule:

$$\frac{d}{dt} \left(\frac{s}{x} \right) = \frac{d}{dt} \left(\frac{h}{10} \right)$$

To find $\frac{ds}{dt}$ (the rate at which the tip of the person's shadow is moving concerning time), we need to solve for it:

$$\frac{1}{x} \frac{ds}{dt} - \frac{s}{x^2} \frac{dx}{dt} = 0$$

Plugging in the given values:

$$\frac{1}{10} \frac{ds}{dt} - \frac{6}{100} (-1.5) = 0$$

Solving for $\frac{ds}{dt}$:

$$\frac{ds}{dt} = \frac{6}{10} \times 1.5 = 0.9 \text{ m/s}$$

So, the tip of the person's shadow is moving away from the pole at a rate of 0.9 m/s when they are 10 meters away from the pole.

2 Real Application

Example 2: The 1945 Trinity test was an implosion-type plutonium bomb with an estimated 20 Kiloton explosive yield. It created a fireball of super-heated gas and plasma up to 100,000,000°C, hotter than the core of the sun. 700 microseconds after the detonation of a 1 Megaton thermonuclear warhead, the nuclear fireball is expanding at a rate of 330,000 m³/s. How fast is the surface area of the fireball expanding when the radius is 160 meters?

Solution:

Given:

$$\begin{aligned} \frac{dV}{dt} &= 330,000 \\ r &= 160 \end{aligned}$$

The equations that relate the variables are the Spherical Volume formula and the Spherical Surface Area formula:

$$\begin{aligned} V &= \frac{4}{3} \pi r^3 \\ A &= 4 \pi r^2 \end{aligned}$$

Differentiating the equations implicitly with respect to time t :

$$\begin{aligned} \frac{dV}{dt} &= 4 \pi r^2 \frac{dr}{dt} \\ \frac{dA}{dt} &= 8 \pi r \frac{dr}{dt} \end{aligned}$$

Now, substitute the known values into the Volume equation and solve for $\frac{dr}{dt}$

$$\frac{dr}{dt} = \frac{330000}{4\pi(160)^2}$$

Finally, find the value of $\frac{dr}{dt}$

$$\frac{dr}{dt} = 1.026$$

Now, substitute the known values into the Surface Area equation and solve for $\frac{dA}{dt}$

$$\frac{dA}{dt} = 8\pi(160)(1.026)$$

Finally, find the value of $\frac{dA}{dt}$

$$\frac{dA}{dt} = 1.33 \times 10^9$$

So, the surface area of the fireball is expanding at a rate of $1.33 \times 10^9 \text{ m}^2/\text{s}$

Minima and Maxima

Aiden Jeong

July 2023

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1 Introduction

Whether it is finding the highest point of a projectile or calculating the quickest path light can take through two different mediums, knowing how to find the extrema of a function is crucial in physics. In this handout we will cover how to find and classify local and global extrema of functions.

2 Critical Points

Suppose we want to find the maximum/minimum value of $f(x)$ on a certain interval. Then we have the following conditions:

Critical Points: The minimum or maximum values of $f(x)$ on the interval $[a, b]$ can only occur at one of the following:

- Points where $f'(x) = 0$
- Points where $f'(x)$ is undefined
- At $x = a$ or $x = b$. That is, the endpoints of the interval.

Intuitively, this should make sense: at a maximum the function must go up and down a “hill” so we should have some point where the slope of the tangent line is zero (condition 1). Sometimes, we have a sharp corner at the top of the hill, and the derivative is undefined (2). Also, the function can just be constantly increasing in which case the maximum is just an endpoint (3).

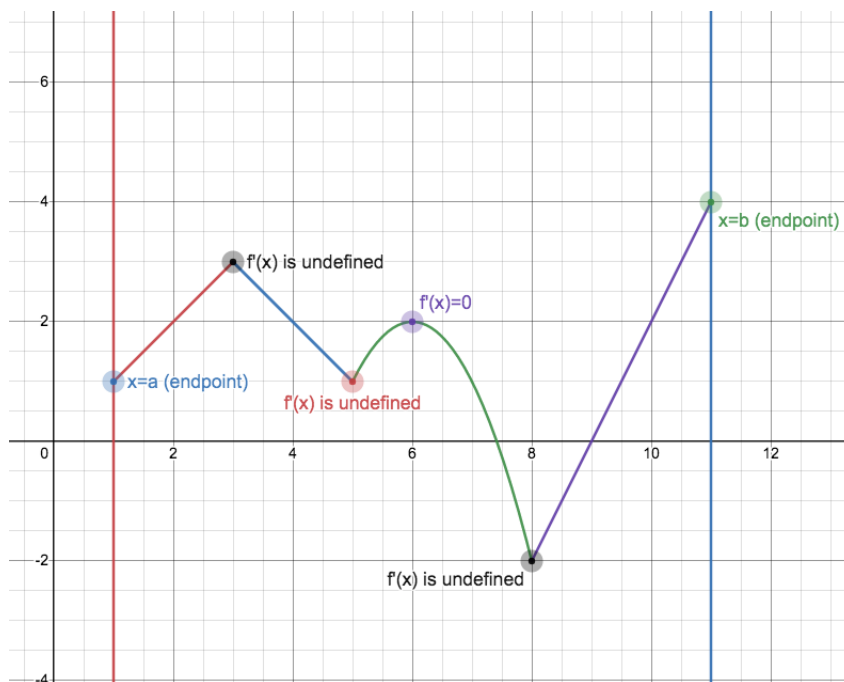


Figure 1: examples in a graph

Notice that these conditions are *not sufficient*. That is, just because an x value satisfies these

conditions does not mean it is a minimum/maximum. Notice this is evident in our graph. Clearly, not all of the marked points are minimums or maximums! Let's try a few examples:

Example 1: Find the critical points of $f(x) = x^2 - 6x + 1$ on the interval $[2, 5]$.

Solution. Let's go through the conditions one by one. We see that $f'(x) = 2x - 6$ so $f'(x) = 0$ at $x = 3$. Clearly, $f'(x)$ is always defined so we don't have to worry about condition two. For condition three, simply take the endpoints $x = 2$ and $x = 5$. Hence, the critical points are at $x = 2, 3, 5$.

Example 2: Find the critical points of the following function on the interval $[1, 12]$:

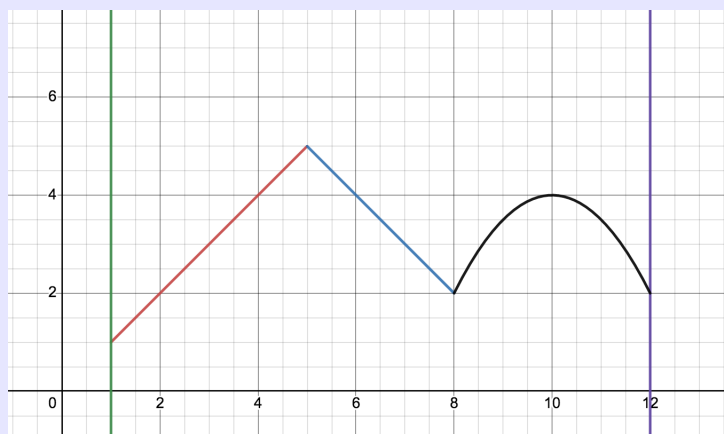


Figure 2: example 2

Solution. We look at each of the conditions:

- For condition one, we have $x = 10$ only.
- For condition two, we have $x = 5, 8$.
- For condition three, we have $x = 1, 12$.

3 Local and Global Extrema

3.1 Definitions

We have a **local minimum** of a function at $x = c$ if $f(c) \leq f(x)$ for all x in *some* open interval (a, b) .

For example, in example 2 we have a local minimum at $x = 8$ since it is the minimum of $f(x)$ on the interval $(6, 10)$. On the other hand, there is no local minimum at $x = 6$ since there is no open interval such that $x = 6$ is the minimum of $f(x)$.

On the other hand, we have a **global minimum** of an interval $[a, b]$ at $x = c$ if $f(c) \leq f(x)$ for all x in $[a, b]$.

For example, for the interval $[7, 10]$ the global minimum is at $x = 8$. Notice global minimums are by definition local minimums. We can define local/global maximums similarly.

3.2 How to Find Global Extrema

Since we know how to find critical points, we can automatically find global extrema. How?

To find the global maximum/minimum in an interval we can evaluate all the critical points and see which one is largest/smallest.

Sometimes, we only have one local minimum/maximum in an interval. Then, this point must automatically be the global minimum/maximum in the interval. Can you see why by sketching an example?

4 Classifying Extrema

We have covered how to identify where potential extrema are. In this section, we find out whether a point is a local maximum or minimum.

4.1 First Derivative Test

First Derivative Test, formal: Let f be a continuous function and suppose c is a real number with $f'(c) = 0$ or $f'(c)$ undefined.

- If there exists an open interval (a, b) containing c such that f is decreasing on (a, c) and increasing on (c, b) then $x = c$ is a local minimum.
- If there exists an open interval (a, b) containing c such that f is increasing on (a, c) and decreasing on (c, b) then $x = c$ is a local maximum.

Here, decreasing and increasing are *not* strict.

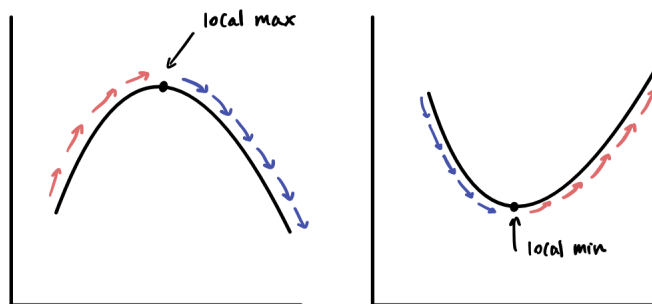


Figure 3: first derivative test, red denotes increasing and blue denotes decreasing

In practice, we just take one point to the left of $x = c$ and one point to the right of $x = c$. If the derivative of the right point is non-positive (≤ 0) and the derivative of the left point is non-negative (≥ 0) we have $x = c$ as a local maximum and vice versa for local minimums.

It is worth noting that the above method does not always work if we take the right and left points too far from $x = c$, so the method is not completely rigorous. In practice, it should work most of the time for simple functions if we take points sufficiently close, though.

Example 3: For the function $f(x) = x^2$ is $x = 0$ a local minimum or maximum?

Solution: First, we know $f'(x) = 2x$. Clearly, $f'(0) = 0$. Take a point to the left of 0, say -1 , and we find $f'(-1) = -2 \leq 0$. Take a point to the right of 0, say 2, and we find $f'(2) = 4 \geq 0$ so $x = 0$ is a local minimum by the first derivative test.

4.2 Second Derivative Test

Second Derivative Test: Suppose $f'(c) = 0$. Then,

- If $f''(c) > 0$ then $x = c$ is a local minimum.
- If $f''(c) < 0$ then $x = c$ is a local maximum.

This should make sense intuitively as concave functions (those with $f''(x) > 0$) are shaped like a “U” so we will have a minimum. Conversely, convex functions (those with $f''(x) < 0$) are shaped like an upside down “U” so we will have a maximum. Also, if $f''(c) = 0$ we cannot make any conclusions about c .

Example 4: Is $x = 3$ a local minimum or maximum for the function $f(x) = -x^2 + 6x$?

Solution: We find $f'(x) = -2x + 6$ so $f'(3) = 0$. Also, $f''(x) = -2$ so $f''(3) = -2 < 0$. Hence, we have $x = 3$ is a local maximum by the second derivative test.

5 Practice Problems

Problem 1

When is $f(x) = 2x^3 - 15x^2 + 36x$ maximized between $x = 0$ and $x = 3.4$?

Problem 2

A baseball is thrown straight up into the air with initial velocity v . Find the maximum height of the baseball. Assume air resistance is negligible.

Problem 3

Walt is selling water. If he sells each gallon at a price of x dollars, Walt expects to sell $10 - x^2$ gallons of water. What is the value of x where Walt's profits are maximized? (Note: x can be any non-negative real number)

Problem 4

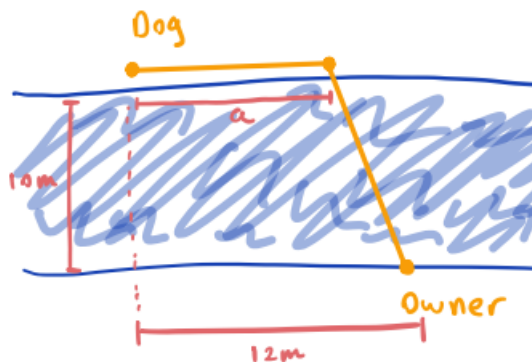


fig 5. problem 4

As shown in the figure, a dog is trying to get back to her owner, who is across the river. The river is 10 meters wide and the dog is 12 meters downriver from her owner. If the dog runs along the shore for a meters at 5 meters per second then swims directly to her owner at 2 meters per second, which value of a minimizes the time spent getting to her owner? (Note: this problem is computationally heavy so the use of a calculator is permitted)

6 Solutions to Practice Problems

Solution 1

Note $f'(x) = 6x^2 - 30x + 36 = 6(x - 3)(x - 2)$. We find the critical points to be $x = 0, 2, 3, 3.4$ as $f'(x)$ is always defined. Testing all these values, we see that we have the largest value at $\boxed{x = 2}$. Note we can see $x = 2$ is a local maximum by the first or second derivative test.

Solution 2

By a basic formula from kinematics, the y coordinate of the baseball can be written as

$$f(t) = vt - \frac{1}{2}gt^2$$

at a given time t , where $g = 9.81 \text{ m/s}^2$. Taking the derivative, we see $f'(t) = v - gt$ so f is maximized at $t = v/g$. This is because we can check $f'(t) > 0$ for $t < v/g$ and $f'(t) < 0$ for $t > v/g$ so $t = v/g$ is an global maximum by the first derivative test. Hence, the maximum height of the baseball is

$$f\left(\frac{v}{g}\right) = \frac{v^2}{g} - \frac{g}{2} \cdot \frac{v^2}{g^2} = \boxed{\frac{v^2}{2g}}$$

Solution 3

Walt's profits can be calculated by the number of gallons of water he sells times the price of each gallon, which is $f(x) = x(10 - x^2)$. Since $f'(x) = 10 - 3x^2$ we have $x = \pm\sqrt{10/3}$ as critical points. Clearly, $x \geq 0$ so we can discard the negative root.

Since $f''(x) = -6x$ we have $f''(\sqrt{10/3}) < 0$ so $x = \sqrt{10/3}$ is a local maximum by the second derivative test. Note only one local extrema point in the interval $[0, \infty)$ so $\boxed{x = \sqrt{10/3}}$ is value of x where Walt's profits are maximized.

Solution 4

First, note a is bounded by 0 and 12. We calculate the time the dog takes to travel the two segments of her path. For the segment on land, she takes $a/5$ seconds. The length of the segment in water is $\sqrt{(12 - a)^2 + 10^2}$ by the Pythagorean theorem, so the time taken is $\frac{1}{2}\sqrt{(12 - a)^2 + 10^2}$ seconds. Hence, it suffices to minimize

$$f(a) = \frac{a}{5} + \frac{\sqrt{(12 - a)^2 + 10^2}}{2} = \frac{a}{5} + \frac{\sqrt{244 - 24a + a^2}}{2}$$

Taking the derivative using Chain Rule,

$$f'(a) = \frac{1}{5} + \frac{-24 + 2a}{4\sqrt{244 - 24a + a^2}}$$

Setting this equal to zero, we have

$$\begin{aligned} \frac{1}{5} &= \frac{24 - 2a}{4\sqrt{244 - 24a + a^2}} \implies 2\sqrt{244 - 24a + a^2} = 5(12 - a) \\ &\implies 4(244 - 24a + a^2) = 25(12 - a)^2 = 25(144 - 24a + a^2) \\ &\implies a \approx 7.64, 16.36 \end{aligned}$$

In the interval $[0, 12]$ the local extrema is at $a = 7.64$. We now must verify that $a = 7.64$ indeed yields a local minimum, which would finish as there is only one local minimum/maximum. We can plug in $a = 7.7$ and $a = 7.6$ and check $f'(7.7) > 0$ and $f'(7.6) < 0$, which implies the desired result by the first derivative test.

Implicit Differentiation

Harsh Ambardekar

July 2023

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1 Introduction

Up to this point, we've differentiated functions by isolating for y and then taking the derivative of the other side. But it's not always easy nor beneficial to solve directly for y . This is where **implicit differentiation** comes in.

Definition: The idea of implicit differentiation is to differentiate both sides initially and use the chain rule. This works because differentiation is an operation like addition or subtraction are. If we add 4 to both sides of an equation, the equation is still true. Likewise, if we differentiate both sides of an equation, the equation is still true.

2 Examples

Example 1: Let's find $\frac{dy}{dx}$ given

$$x^2 + y^2 = 1$$

Solution:

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(1)$$

$$2x + 2y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-x}{y}$$

In the third line, we used the chain rule to differentiate y^2 ; y is a function of x so we must multiply $2y$ by $\frac{dy}{dx}$. Note that you may see $\frac{dy}{dx}$ written as y' . Both mean the same thing.

Example 2: Find y' given $x^2 + xy + y^3 = 0$.

Solution: We have to be careful to use both the chain rule and the product rule here, since the middle term is xy , which is a product of two functions.

$$\frac{d}{dx}(x^2 + xy + y^3) = \frac{d}{dx}(0)$$

$$2x + (x \cdot y' + 1 \cdot y) + 3y^2 y' = 0$$

Now, we solve for y' :

$$y'(x + 3y^2) = -2x - y$$

$$y' = \frac{-2x - y}{x + 3y^2}$$

3 Practice

3.1 Problems

Problem 1: Given $2xy + x^3 - 3y^2 = 5$, find $\frac{dy}{dx}$.

Problem 2: Given $x^3y + y^2 - x^2 = 5$, find the slope of the tangent line at the point (2,1).

Problem 3: If $\frac{1}{x} + \frac{1}{y} = \frac{1}{12}$, find $\frac{dy}{dx}$.

3.2 Answers

Answer 1: $\frac{-2y-3x^2}{2x-6y}$

Answer 2: $-\frac{4}{5}$

Answer 3: $-\frac{y^2}{x^2}$

You can find many more practice problems online (Khan Academy, Paul's Online Math Notes, Organic Chemistry Tutor, etc).

Chain Rule

Logan Reich

August 2023

1 Introduction

How do we take the derivative of $\sin(3x)$? We know how to take the derivative of $\sin(x)$, so it may seem like $\sin(3x)$ would just be $\cos(3x)$ - but it's not. Instead, we have to apply the chain rule. The chain rule is a simple yet incredibly important rule in calculus that allows us to take the derivative of more advanced functions more easily.

At its most basic level, the **chain rule** states that if:

$$f(x) = g(h(x))$$

then:

$$f'(x) = \frac{df(x)}{dx} = g'(h(x))h'(x)$$

In words, this means that the derivative of a function f that equals g of h of x , where h and g are any function, then the derivative of f , f' of x , equals g' prime of h of x , or $h(x)$ plugged into the derivative of the function g times the derivative of the function h evaluated at x . This may initially seem strange or abstract, and makes more sense with an example.

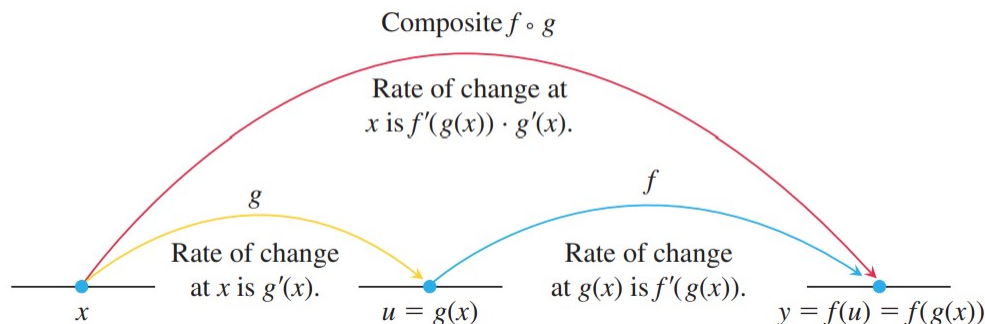


Figure 1: A Depiction of the Chain Rule (Thomas Calculus)

Example 1: What is the derivative of $\sin(3x)$?

Solution: Let $f(x) = \sin(3x) = g(h(x))$ where $g(x) = \sin(x)$ and $h(x) = 3x$ then:

$$f'(x) = \frac{df(x)}{dx} = g'(h(x))h'(x) = \cos(3x)3 = 3\cos(3x)$$

since the derivative of $\sin$ is $\cos$ and the derivative of $3x$ is 3 .

This principle can be extended to even more complex functions, such as in the below example.

Example 2: What is the derivative of $\sin(\sin(\sin(x)))$?

Solution: Let $f(x) = \sin(\sin(\sin(x))) = g(h(x)) = g(p(q(x)))$ where $g(x) = \sin(x)$ $h(x) = p(q(x)) = \sin(\sin(x))$ $p(x) = \sin(x)$ $q(x) = \sin(x)$ so:

$$\begin{aligned} f'(x) &= \frac{df(x)}{dx} = g'(h(x))h'(x) = \cos(\sin(\sin(x)))h'(x) \\ &= \cos(\sin(\sin(x)))p'(q(x))q'(x) = \cos(\sin(\sin(x)))\cos(\sin(x))\cos(x) \end{aligned}$$

And that's the chain rule for you!

Now, time for a quick proof of the chain rule!

We can express the derivative of $f(g(x))$ as:

$$(f(g(x)))' = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a}$$

Noting that $g(x)$ does not equal $g(a)$ for x near a unless the function is a straight, horizontal line in which case we do not need the chain rule:

$$= \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \frac{g(x) - g(a)}{x - a}$$

Since the limit of the product is the product of the limits, and then using the definition of a derivative, this equals our now-familiar $f'(g(x))g'(x)$

Now it's time for some examples! A rocket, like the one shown below, has a position function

$$r(t) = 5t + 0.5 \sin(3t^2 + 4t)$$



Figure 2: A Rocket (Wikipedia)

What is its velocity? Ans: $v(t) = 5 + 0.5 \cos(3t^2 + 4t)(6t + 4)(6)$

Take the derivative of the following functions.

$$4(5x + 34)^2$$

Ans: $8(5x + 34)5$

$$5(4x^2 + 6x + 38)^2$$

Ans: $10(4x^2 + 6x + 38)(8x + 6)(8)$

$$6(2x + 3)^3$$

Ans: $18(2x + 3)^2(2)$

$$\sin(4x^2 + 3)$$

Ans: $\cos(4x^2 + 3)(8x)(8)$

$$\tan(3x + 2)$$

Ans: $\sec^2(3x + 2)(3)$

$$\left(\frac{4x - 3}{2x + 1}\right)^2$$

Ans: $2\left(\frac{4x-3}{2x+1}\right)\left(\frac{(2x+1)(4)-(4x-3)(2)}{(2x+1)^2}\right)$

$$\sin(\cos(4x^2))$$

Ans: $\cos(\cos(4x^2))(-\sin(4x^2))(8x)$

$$\sin(10x)$$

Ans: $\cos(10x)(10)$

$$\tan(3x^2 + 4x)$$

Ans: $\sec^2(3x^2 + 4x)(6x + 4)(6)$

$$3e^{3x}$$

Ans: $3e^{3x}3$

$$5e^{16x^2+4x}$$

Ans: $5e^{16x^2+4x}(32x+4)(32)$

$$\sin(5e^{16x^2+4x})$$

Ans: $\cos(5e^{16x^2+4x})5e^{16x^2+4x}(32x+4)(32)$

$$\sin(\cos(\sin(\cos(10x^2+1))))$$

Ans: $\cos(\cos(\sin(\cos(10x^2+1))))(-\sin(\sin(\cos(10x^2+1))))\cos(\cos(10x^2+1))(-\sin(10x^2+1))(20x)(20)$

Differentiation Rules

Ashwin Narayanan

August 2023

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1 Trigonometric Derivatives

Trigonometric derivatives are mostly formulaic, and require an understanding of trig derivative rules in order to solve differentiation problems involving trig functions. Here are the derivatives of common trigonometric functions: The derivative of the sine function, denoted as $\sin(x)$, is the cosine function, denoted as $\cos(x)$:

$$\frac{d}{dx} \sin(x) = \cos(x)$$

The derivative of the cosine function, denoted as $\cos(x)$, is the negative sine function, denoted as $-\sin(x)$:

$$\frac{d}{dx} \cos(x) = -\sin(x)$$

The derivative of the tangent function, denoted as $\tan(x)$, is the secant squared function, denoted as $\sec^2(x)$:

$$\frac{d}{dx} \tan(x) = \sec^2(x)$$

The derivative of the cotangent function, denoted as $\cot(x)$, is the negative cosecant squared function, denoted as $-\csc^2(x)$:

$$\frac{d}{dx} \cot(x) = -\csc^2(x)$$

The derivative of the secant function, denoted as $\sec(x)$, is the secant function itself multiplied by the tangent function, denoted as $\sec(x) \tan(x)$:

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

The derivative of the cosecant function, denoted as $\csc(x)$, is the negative cosecant function itself multiplied by the cotangent function, denoted as $-\csc(x) \cot(x)$:

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x)$$

Example 1: Consider the trigonometric function $f(x) = -\sin(x) - \cos(x)$. Apply trig differentiation rules to compute the derivative.

Solution: Step 1: Using the rule $\frac{d}{dx}(\sin(x)) = \cos(x)$, the derivative of $-\sin(x)$ is $-\cos(x)$. Step 2: Using the rule $\frac{d}{dx}(\cos(x)) = -\sin(x)$, the derivative of $-\cos(x)$ is $\sin(x)$. Now, we have found the derivatives of the individual terms:

$$\frac{d}{dx}(-\sin(x)) = -\cos(x)$$

and

$$\frac{d}{dx}(-\cos(x)) = \sin(x)$$

So, the derivative of $f(x) = -\sin(x) - \cos(x)$ is the sum of the derivatives:

$$\frac{d}{dx}(-\sin(x) - \cos(x)) = -\cos(x) + \sin(x)$$

2 Power Rule

The power rule is a fundamental rule that allows us to differentiate functions of the form $f(x) = x^n$, where n is a constant exponent.

The **Power Rule** states that if $f(x) = x^n$, then the derivative of $f(x)$ with respect to x is given by:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

In this equation, $\frac{d}{dx}$ represents the derivative operator, and nx^{n-1} is the result of applying the power rule. It shows that when we differentiate a term with an exponent n , we bring down the exponent as the new coefficient and reduce the exponent by 1. For example, if we have $g(x) = x^3$, the power rule tells us that the derivative of $g(x)$ is $g'(x) = 3x^{3-1} = 3x^2$. So, the derivative of $g(x) = x^3$ is $g'(x) = 3x^2$.

Example 2: Consider the function $g(x) = x^{-2}$. Apply the power rule to calculate the derivative.

Solution: Using the power rule, we find: $g'(x) = -2x^{-2-1} = -2x^{-3} = \frac{-2}{x^3}$. Therefore, the derivative of $g(x) = x^{-2}$ is $g'(x) = \frac{-2}{x^3}$.

3 Product Rule

The product rule allows us to find the derivative of a product of two functions and is a valuable tool that allows us to find the derivative of more complex expressions.

The **Product Rule** states that if we have two functions, $f(x)$ and $g(x)$, then the derivative of their product, $h(x) = f(x) \cdot g(x)$, is given by:

$$h'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

In this equation, $f'(x)$ represents the derivative of $f(x)$, and $g'(x)$ represents the derivative of $g(x)$. The product rule tells us that to find the derivative of a product, we take the derivative of the first function and multiply it by the second function, then add the product of the first function and the derivative of the second function. For example, let's consider $f(x) = x^2$ and $g(x) = \sin(x)$. Applying the product rule, we find:

$$h'(x) = (x^2)' \cdot \sin(x) + x^2 \cdot (\sin(x))'$$

Simplifying further:

$$h'(x) = 2x \cdot \sin(x) + x^2 \cdot \cos(x)$$

Therefore, the derivative of $h(x) = x^2 \cdot \sin(x)$ is $h'(x) = 2x \cdot \sin(x) + x^2 \cdot \cos(x)$.

I use a little mnemonic to remember the product rule that goes something like this: "First d -second $\cdot$ second d -first", the first and second representing $f(x)$ and $g(x)$, and the d prefix representing which function must be differentiated

Example 3: Consider the function $g(x) = (2x + 6)(\sin(x))$. Apply the product rule to calculate the derivative.

Solution: Recall the product rule formula:

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

where $f(x)$ and $g(x)$ are two differentiable functions. Step 1: identify the functions $f(x)$ and $g(x)$:

$$f(x) = 2x + 6$$

$$g(x) = \sin(x)$$

Step 2: Find the derivatives $f'(x)$ and $g'(x)$:

$$f'(x) = \frac{d}{dx}(2x + 6) = 2$$

$$g'(x) = \frac{d}{dx}(\sin(x)) = \cos(x)$$

Step 3: Apply the product rule:

$$g'(x) = f'(x)g(x) + f(x)g'(x) = 2\sin(x) + (2x + 6)\cos(x)$$

The derivative of $g(x) = (2x + 6)\sin(x)$ with respect to x is $g'(x) = 2\sin(x) + (2x + 6)\cos(x)$.

4 Quotient Rule

The **Quotient Rule** states that if we have two functions, $f(x)$ and $g(x)$, then the derivative of their quotient, $h(x) = \frac{f(x)}{g(x)}$, is given by:

$$h'(x) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$$

In this equation, $f'(x)$ represents the derivative of $f(x)$, and $g'(x)$ represents the derivative of $g(x)$.

The quotient rule tells us that to find the derivative of a quotient, we take the derivative of the numerator and multiply it by the denominator, then subtract the product of the numerator and the derivative of the denominator, all divided by the square of the denominator. For example, let's consider $f(x) = \sin(x)$ and $g(x) = x^2$. Applying the quotient rule, we find:

$$h'(x) = \frac{(\sin(x))' \cdot x^2 - \sin(x) \cdot (x^2)'}{(x^2)^2}$$

Simplifying further:

$$h'(x) = \frac{\cos(x) \cdot x^2 - \sin(x) \cdot 2x}{x^4}$$

Therefore, the derivative of $h(x) = \frac{\sin(x)}{x^2}$ is $h'(x) = \frac{\cos(x) \cdot x^2 - \sin(x) \cdot 2x}{x^4}$.

The quotient rule is particularly useful when dealing with functions that cannot be easily differentiated using other basic rules such as the power rule or the product rule.

Example 4: Consider the function $h(x) = \frac{x^3}{\cos(x)}$. Apply the quotient rule to calculate the derivative.

Solution: To find the derivative $h'(x)$, we'll use the quotient rule:

$$h'(x) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$$

First, we'll identify $f(x)$ and $g(x)$ and plug them into the quotient rule formula:

$$f(x) = x^3$$

$$g(x) = \cos(x)$$

$$h'(x) = \frac{(x^3)' \cos(x) - x^3 (\cos(x))'}{(\cos(x))^2}$$

Now, let's calculate the derivatives of the individual functions:

$$(x^3)' = 3x^2$$

$$\cos(x)' = -\sin(x)$$

Substituting these derivatives into the quotient rule:

$$h'(x) = \frac{3x^2 \cdot \cos(x) - x^3 \cdot (-\sin(x))}{(\cos(x))^2}$$

Simplify further:

$$h'(x) = \frac{3x^2 \cos(x) + x^3 \sin(x)}{\cos^2(x)}$$

So, the derivative of $h(x) = \frac{x^3}{\cos(x)}$ is $h'(x) = \frac{3x^2 \cos(x) + x^3 \sin(x)}{\cos^2(x)}$.

5 Multiple Derivatives

The double derivative, also known as the second derivative, is a fundamental idea in calculus that represents the rate of change of a function's derivative. In other words, it tells us how the slope of a function is changing as we move along the x-axis. The double derivative provides information about the curvature and concavity of a function.

Let's consider a function $f(x)$. The first derivative of $f(x)$ with respect to x is denoted as $f'(x)$, and the second derivative is denoted as $f''(x)$. Mathematically, the **Second Derivative** is the derivative of the first derivative:

$$f''(x) = \frac{d}{dx} \left(\frac{d}{dx} f(x) \right) = \frac{d^2}{dx^2}$$

Multiple differentiation can keep occurring like so:

$$f'''(x) = \frac{d}{dx} \left(\frac{d}{dx} \left(\frac{d}{dx} f(x) \right) \right) = \frac{d^3}{dx^3}$$

Which represents the third derivative, that is the rate of change of the rate of change of the rate of change.

Now, let's look at an example to illustrate the concept: Let's consider the function

$$f(x) = 3x^3 + 2x^2 - 5x + 1$$

Step 1: Find the first derivative $f'(x)$:

$$f'(x) = \frac{d}{dx}(3x^3 + 2x^2 - 5x + 1)$$

$$f'(x) = 9x^2 + 4x - 5$$

Step 2: Find the second derivative $f''(x)$:

$$f''(x) = \frac{d}{dx}(9x^2 + 4x - 5)$$

$$f''(x) = 18x + 4$$

So, the second derivative of the function $f(x) = 3x^3 + 2x^2 - 5x + 1$ is $f''(x) = 18x + 4$. In this example, the second derivative $f''(x)$ gives us information about how the slope of the original function $f(x)$ is changing at different points on the graph.

A positive value of $f''(x)$ indicates the function is **concave up** ($\cup$), while a negative value indicates the function is **concave down** ($\cap$).

Example 5: Compute the third derivative of the following function: $f(x) = \sin(x)$.

Solution:

Step 1: Evaluate the first derivative:

$$f'(x) = \frac{d}{dx}(\sin(x)) = \cos(x)$$

Step 2: Evaluate the second derivative:

$$f''(x) = \frac{d^2}{dx^2}(\sin(x)) = \frac{d}{dx}(\cos(x)) = -\sin(x)$$

Step 3: Evaluate the third derivative:

$$f'''(x) = \frac{d^3}{dx^3}(\sin(x)) = \frac{d}{dx}(-\sin(x)) = -\cos(x)$$

$$f'''(x) = -\cos(x)$$

Introduction to Derivatives

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1 Introduction

The **derivative** is a fundamental concept in calculus that measures the rate at which a function changes with respect to its independent variable. It provides information about how a function behaves locally at or near a particular point.

1.1 Slope

To understand derivatives, let's start with the concept of **slope**. In basic geometry, the slope of a straight line represents how steep or flat the line is. It tells us how the y -coordinate changes as the x -coordinate changes. The slope is defined as the ratio of the vertical change (rise) to the horizontal change (run) between any two points on the line, as represented by the following slope formula:

$$\frac{y_2 - y_1}{x_2 - x_1}$$

representing change in y over change in x . The derivative is essentially this change in y over x , or slope.

1.2 Tangent Lines

Now, let's introduce **tangent lines**. A tangent line is a line that touches a curve at a specific point without crossing through it. The tangent line shares the same slope as the curve at that point. The derivative of a function at a particular point gives us the slope of the tangent line to the graph of the function at that point. It tells us how fast the function is changing at that specific location. To visualize this, imagine a curve representing a function on a graph. At a given point on the curve, we can draw a straight line that barely touches the curve without crossing it. This tangent line captures the instantaneous rate of change of the function at that point.

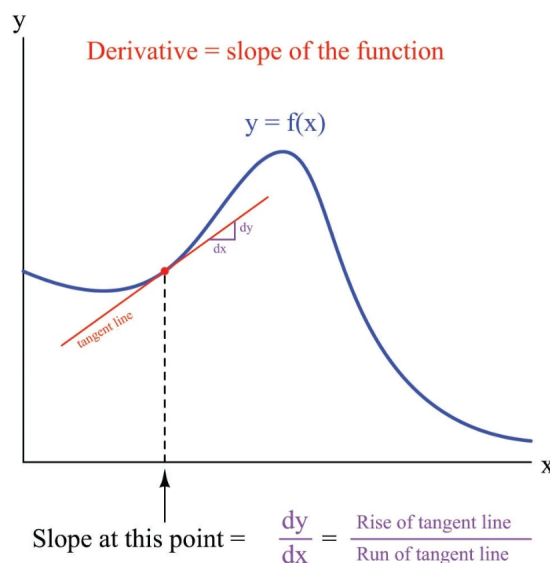


Figure 1: Example of a tangent line and respective derivative. The common derivative operator $\frac{dy}{dx}$ represents the change in y over the change in x

Let's say you have a simple function,

$$f(x) = 3x^2$$

The derivative of this function with respect to x , $f'(x)$, tells you how fast $f(x)$ is changing with respect to x . Derivatives capture essential information about the behavior of functions. It tells us whether a function is increasing or decreasing, the slope of the tangent line at a point, and provides insights into the shape and features of the function.

Example 1: What is the derivative of the function $f(x) = 2x + 1$ at the point $(0, 1)$?

Solution: We know that the derivative has the same meaning as slope, and the slope of the function throughout all values of x is 2. Therefore, the derivative at the point $(0, 1)$ is 2.

2 Definition of the Derivative

To define the derivative, we use the concept of a **limit**. Let's consider a function $f(x)$ and a specific point a on the function.

The derivative of $f(x)$ at the point a , denoted as $f'(a)$ or df/dx evaluated at $x = a$, is given by the following limit:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Here, h represents a small change in the x -coordinate from a . We want to see how the function $f(x)$ changes as we take smaller and smaller values of h . Taking the limit as h approaches zero gives us the precise instantaneous rate of change. To calculate the derivative, we solve the **difference quotient**:

$$\frac{f(a+h) - f(a)}{h}$$

which represents the average rate of change of the function over the interval $[a, a+h]$. Then, as h approaches zero, the limit determines the precise rate of change at the point a . In other words, we're zooming in on the point a and examining how the function behaves within an infinitesimally small interval around that point.

If this limit exists, we say the function is **differentiable** at the point a , and the limit value represents the derivative of $f(x)$ at a . It's important to note that **not all functions have derivatives at every point**. Some functions may have discontinuities or sharp corners that prevent the existence of a derivative.

2.1 Discontinuities

To recognize when a derivative does not exist for a function, there are a few key scenarios to consider. Here are some common cases:

- **Discontinuities:** Simply put, a function is *continuous* if you can draw it without lifting your pencil. If you ever have to lift your pencil, then that point is called a *discontinuity*. Discontinuities come in 3 main types:
 - **Removable:** A removable discontinuity occurs when the function *looks* continuous, except at that point, there is a hole in the graph.
 - **Jump:** A jump discontinuity occurs when the graph appears to “jump” from one curve to another
 - **Infinite:** An infinite discontinuity occurs where the function approaches infinity as it

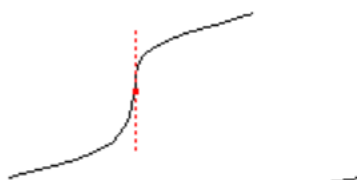
If a function has a discontinuity, then the derivative does not exist at that point. Even if the derivative approaches the same value as you take the limit from either side of the discontinuity, since the function is not continuous at that specific point, it cannot have a derivative. However, as we will see in the next two examples, just because a function is continuous that does NOT mean it is differentiable.

- **Corner or cusp points:** At a corner or cusp point, the function abruptly changes direction without having a well-defined tangent line. In such cases, the derivative does not exist at that point.
- **Vertical tangent lines:** If the slope of the tangent line becomes infinite (vertical), the derivative is said to be undefined at that point. This typically occurs when the function has a vertical tangent, where the slope increases or decreases without bound.

Corner or Cusp:



Vertical Tangent:



Point of Discontinuity:



Figure 2: Examples of functions where the derivative does not exist at a certain point

Example 2: Use limits and the definition of the derivative to find the derivative function f' for each function f .

(a) $f(x) = 5x^3 + 3x$

(b) $f(x) = x^{-2} + 4$

3 Worked Problems

Solution: (a) To find the derivative function, we need to compute the limit as h approaches 0 of the difference quotient:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Let's substitute the function $f(x) = 5x^3 + 3x$ into the difference quotient:

$$f'(x) = \lim_{h \rightarrow 0} \frac{5(x+h)^3 + 3(x+h) - (5x^3 + 3x)}{h}$$

Expanding the cube and simplifying, we have:

$$f'(x) = \lim_{h \rightarrow 0} \frac{5(x^3 + 3x^2h + 3xh^2 + h^3) + 3(x+h) - (5x^3 + 3x)}{h}$$

Let's distribute the 5 to each term inside the parentheses:

$$f'(x) = \lim_{h \rightarrow 0} \frac{5x^3 + 15x^2h + 15xh^2 + 5h^3 + 3x + 3h - 5x^3 - 3x}{h}$$

Notice that the $5x^3$ and $-5x^3$ terms cancel out, as well as the $3x$ and $-3x$ terms:

$$f'(x) = \lim_{h \rightarrow 0} \frac{15x^2h + 15xh^2 + 5h^3 + 3h}{h}$$

We can factor out an h from the numerator:

$$f'(x) = \lim_{h \rightarrow 0} \frac{h(15x^2 + 15xh + 5h^2 + 3)}{h}$$

Canceling out the h terms:

$$f'(x) = \lim_{h \rightarrow 0} 15x^2 + 15xh + 5h^2 + 3$$

Now, take the limit as h approaches 0. Notice that all the terms with h will vanish:

$$f'(x) = 15x^2 + 3$$

So, the derivative of the function $f(x) = 5x^3 + 3x$ is $f'(x) = 15x^2 + 3$.

(b) Let's start with the definition itself:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Let's substitute the function $f(x) = x^{-2} + 4$ into the definition of the derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^{-2} + 4 - (x^{-2} + 4)}{h}$$

Now, let's simplify the expression inside the limit:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^{-2} - x^{-2}}{h}$$

To simplify further, we can rewrite the expression as:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\left(\frac{1}{(x+h)^2} - \frac{1}{x^2} \right)}{h}$$

Next, we'll combine the fractions by finding a common denominator:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\left(\frac{x^2 - (x+h)^2}{x^2(x+h)^2} \right)}{h}$$

Expanding $(x+h)^2$ gives:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\left(\frac{x^2 - (x^2 + 2xh + h^2)}{x^2(x+h)^2} \right)}{h}$$

Simplifying the numerator:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\left(\frac{-2xh - h^2}{x^2(x+h)^2} \right)}{h}$$

Now, we can cancel out the h in the numerator and denominator:

$$f'(x) = \lim_{h \rightarrow 0} \frac{-2x - h}{x^2(x+h)^2}$$

Taking the limit as h approaches 0:

$$f'(x) = \frac{-2x}{x^2(x)^2}$$

Simplifying further:

$$f'(x) = \frac{-2}{x^3}$$

Hence, the derivative of $f(x) = x^{-2} + 4$ is $f'(x) = -\frac{2}{x^3}$.